Exercise list R2c — Financial planning and forecasting for valuation

Worked solutions

Statements are in `Listas/lista-r2c-financial-planning-forecasting.md`; each exercise is cited by number and printed page and its full text is reproduced right before its solution. Short answers: `lista-r2c-financial-planning-forecasting-solutions-short.pdf`. Editions are the local copies: Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011), Ch. 18; Higgins = Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed., Ch. 3–4; Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed., Cap. 4 ("NAF" = *necessidade de aporte financeiro* = EFN). Amounts carry the units of the statement (dollars, $ thousands or $ millions). Numbers marked "illustration" are not part of the statement.

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1. Cost of goods sold by the percent of sales method — Berk 18.1, p. 558

Statement.

Source: Berk, Ch. 18, Exercise 18.1, p. 558 (PDF 589) · Answer in book: no

1. Your company has sales of $100,000 this year and cost of goods sold of $72,000. You forecast sales to increase to $110,000 next year. Using the percent of sales method, forecast next year's cost of goods sold.

What is asked. Next year's cost of goods sold (COGS) when sales rise from $100,000 to $110,000, assuming COGS keeps this year's ratio to sales.

Approach. The percent of sales method (Berk §18.2) assumes that an item that moves with activity keeps the same fraction of sales. So we measure the fraction today and apply it to forecast sales. This is the first line of any pro forma income statement.

Notation.

- $S_0 = 100{,}000$, $S_1 = 110{,}000$: sales this year and next year ($).
- $C_0 = 72{,}000$: COGS this year ($); $c = C_0/S_0$: COGS as a fraction of sales; C_1: forecast COGS.

Step 1 — Measure the ratio. We divide this year's COGS by this year's sales:

$$c = \frac{C_0}{S_0} = \frac{72{,}000}{100{,}000} = 0.72. \tag{1}$$

Step 2 — Apply it to forecast sales. The ratio is assumed constant, so

$$C_1 = c\,S_1 = (0.72)(110{,}000) = 79{,}200. \tag{2}$$

Answer . Forecast COGS = $79,200 (72% of $110,000).

Intuition. Holding COGS at 72% of sales assumes the gross margin (28%) is stable: each extra dollar of sales brings 72 cents of extra cost. It is a good first guess for a mature business; a firm with fixed production costs would see COGS grow more slowly than sales.

Common mistakes.

- Growing COGS by a different rate than sales (e.g. by last year's COGS growth) — the method ties every proportional item to the *sales* forecast.
- Applying the ratio to this year's sales instead of next year's.

Ref: Berk §18.2, "Percent of Sales Method", pp. 537–538; Example 18.1, p. 539.

2. Reading external financing required — Higgins 3.1 and 3.4, p. 103

Statement.

Source: Higgins, Ch. 3, Exercises 3.1 and 3.4, p. 103 (PDF 124) · Answer in book: yes for 3.1, p. 405; no for 3.4

1. Suppose you constructed a pro forma balance sheet for a company and the estimate for external financing required was negative. How would you interpret this result?
2. Answer true or false to each of the following. Briefly explain your reasoning for each answer.

a. All else equal, increasing the projected amount of accounts receivable in a financial forecast will increase external funding required.

b. Estimates of external funding required based on cash flow forecasts are usually higher than estimates based on pro forma financial statements.

c. An annual financial forecast for 2022 showing no external funding required assures a company that no cash shortfalls are likely to occur during 2022.

What is asked. (3.1) What a *negative* external financing required means. (3.4a) Whether a larger receivables forecast raises external funding required. (3.4b) Whether cash flow forecasts usually show larger needs than pro forma statements. (3.4c) Whether an annual forecast with no external need rules out cash shortfalls during the year.

Approach. In a pro forma balance sheet, external funding required (EFN) is the *plug*: the amount that makes assets equal liabilities plus equity. Every question follows from the identity that defines it and from the date at which the identity is evaluated.

Notation.

- TA: projected total assets; L: projected liabilities (without new financing); E: projected equity.
- $EFN = TA - (L + E)$: external funding required at the forecast date.

Step 1 — The defining identity. By construction,

$$EFN = TA - (L + E). \tag{3}$$

A positive value means projected uses (assets) exceed projected sources: money must be raised.

Step 2 — (3.1) Negative EFN. If $EFN < 0$, sources exceed uses: the company will generate more cash than its plan absorbs. Illustration: projected assets 1,000 with cash at its minimum of 100, and $L + E = 1{,}040$; then $EFN = 1{,}000 - 1{,}040 = -40$. Setting the plug to zero and adding the 40 to cash gives cash $= 100 + 40 = 140$ and assets $= 900 + 140 = 1{,}040 = L + E$: the balance sheet balances.

Answer (3.1). A negative EFN is a projected cash *surplus* above the desired minimum: cash (or debt repayment, dividends, buybacks) absorbs it.

Step 3 — (3.4a) Receivables. Receivables are part of TA in (3). Holding L and E fixed, $\partial EFN/\partial AR = 1$: in the illustration, raising receivables by 50 moves EFN from -40 to $-40+50 = 10$.

Answer (3.4a). True: every extra dollar tied up in receivables must be financed.

Step 4 — (3.4b) Cash flow forecasts. A cash flow forecast and a pro forma balance sheet rearrange the same accounting identities; Higgins' chapter summary states that cash flow forecasts "yield the same need for external funding as a pro forma projection, given the same assumptions."

Answer (3.4b). False: with the same assumptions and no errors, the two estimates are identical.

Step 5 — (3.4c) Intra-year needs. (3) is evaluated on one date (December 31). A seasonal firm can build inventory and receivables in mid-year and need financing then, even if by year-end the cash has come back.

Answer (3.4c). False: an annual forecast says nothing about needs between forecast dates; monthly cash budgets or quarterly pro formas are needed for that.

Intuition. EFN is a balancing item, not a forecast of performance. Its sign tells you whether the plan throws off or absorbs cash *at the forecast date*, and anything that adds to assets raises it one-for-one.

Common mistakes.

- Reading a negative EFN as "the forecast is wrong"; it is a cash surplus.
- Believing different forecasting methods give different needs; differences signal an error.
- Using a year-end forecast to manage a seasonal business.

Book answer. 3.1 matches the book (p. 405): "A negative value implies that the company has excess cash above its desired minimum", confirmed by adding it to cash. 3.4: no answer in the book.

Ref: Higgins Ch. 3, "Pro Forma Statements" and chapter summary, pp. 80–102.

3. Corporação Felipe: the plug variable — Ross 4.1 and 4.2, p. 117

Statement.

Source: Ross, Cap. 4, Exercises 4.1 and 4.2, p. 117 (PDF 141) · Answer in book: no (selected answers in the web-only Apêndice C)

1. **Demonstrações projetadas** [OA1] Considere as seguintes demonstrações contábeis simplificadas da Corporação Felipe (assumindo que não há imposto de renda):

Demonstração de resultados		Balanço patrimonial			
Vendas	\$23.000	Ativo	\$15.800	Dívida	\$ 5.200
Custos	16.700			Patrimônio líquido	10.600
Lucro líquido	\$ 6.300	Total	\$15.800	Total	\$15.800

A Felipe previu um aumento de 15% nas vendas. Também previu que cada item do balanço patrimonial crescerá 15%. Construa as demonstrações projetadas e deixe-as equilibradas. Qual é a variável de fechamento?

2. **Demonstrações projetadas e NAF** [OA1, OA2] Na questão anterior, assuma que a Felipe distribua metade do lucro líquido como dividendos. Os custos e o ativo variam com as vendas, mas a dívida e o patrimônio líquido não. Prepare as demonstrações projetadas e determine o aporte financeiro necessário.

What is asked. (4.1) Pro forma statements when sales and *every* balance sheet item grow 15%, and which variable closes the balance sheet. (4.2) Same forecast, but half of net income is paid as dividends and debt and equity do not grow with sales: find the external financing needed (NAF).

Approach. A pro forma model has one degree of freedom too many once we fix sales growth, the percentages, and the financing; one item must be the *variable de fechamento* (plug). In 4.1 the balance sheet is fully specified, so dividends become the plug; in 4.2 the payout is fixed, so the plug moves to external financing.

Notation.

- $g = 15\%$: sales growth; S, C, NI: sales, costs, net income (no taxes).
- A, D, E: assets, debt, equity; Div: dividends; EFN (NAF): external financing needed.

Step 1 — Pro forma income statement (both parts). Costs vary with sales:

$$S_1 = 23{,}000(1.15) = 26{,}450, \qquad C_1 = 16{,}700(1.15) = 19{,}205, \tag{4}$$

$$NI_1 = S_1 - C_1 = 26{,}450 - 19{,}205 = 7{,}245. \tag{5}$$

Step 2 — (4.1) Balance sheet grown 15%. Every item grows 15%: $A_1 = 15{,}800(1.15) = 18{,}170$, $D_1 = 5{,}200(1.15) = 5{,}980$, $E_1 = 10{,}600(1.15) = 12{,}190$, and $5{,}980 + 12{,}190 = 18{,}170$, so it balances.

Step 3 — (4.1) Find the plug. Equity can only change through retained earnings (no share issues), so $E_1 = E_0 + NI_1 - Div_1$. Solving for dividends,

$$Div_1 = NI_1 - (E_1 - E_0) = 7{,}245 - (12{,}190 - 10{,}600) = 7{,}245 - 1{,}590 = 5{,}655. \tag{6}$$

If no dividend were paid, equity would be $10{,}600 + 7{,}245 = 17{,}845$ and the balance sheet would not balance; the dividend of 5,655 is what makes it do so.

Answer (4.1). Sales 26,450, costs 19,205, net income 7,245; assets 18,170 = debt 5,980 + equity 12,190. The plug (*variável de fechamento*) is the dividend, \$5,655.

Step 4 — (4.2) Fixed payout, fixed debt. Dividends are $0.5(7{,}245) = 3{,}622.5$, so retained earnings add 3,622.5 and

$$E_1 = 10{,}600 + 3{,}622.5 = 14{,}222.5, \qquad D_1 = 5{,}200,$$

$$D_1 + E_1 = 5{,}200 + 14{,}222.5 = 19{,}422.5,$$

$$EFN = A_1 - (D_1 + E_1) = 18{,}170 - 19{,}422.5 = -1{,}252.5. \tag{7}$$

Answer (4.2). NAF = −$1,252.5: a surplus. The firm can pay down debt, add to cash, or raise the dividend by that amount.

Intuition. Once growth and the asset intensity are fixed, *something* must give: dividends, borrowing or new equity. Ross' point is that the pro forma does not tell you which; the manager chooses the plug, and the model reports the consequence.

Common mistakes.

- Growing equity by 15% *and* adding net income: equity changes only through retained earnings.
- Reporting a negative NAF as zero; the sign carries information (excess funds).

Ref: Ross §4.3, "O modelo da porcentagem de vendas", pp. 99–104; Exemplo pp. 97–98.

4. Internal and sustainable growth rates — Berk 18.16, p. 561

Statement.

Source: Berk, Ch. 18, Exercise 18.16, p. 561 (PDF 592) · Answer in book: no

16. Using the information in the table below, calculate this company's:

Net Income	50,000
Beginning Total Assets	400,000
Beginning Stockholders' Equity	250,000
Payout Ratio	0%

a. Internal growth rate.

b. Sustainable growth rate.

c. Sustainable growth rate if it pays out 40% of its net income as a dividend.

Context needed to solve it *From Berk §18.4, "Sustainable Growth Rate and External Financing", p. 548 (PDF 579):*

Internal Growth Rate Formula. Both these benchmark growth rates are aimed at identifying how much growth a firm can support based on its existing net income. For a firm that does not pay any dividends, its internal growth rate is its return on assets, because that tells us how fast it could grow its assets using only its net income. If the firm pays some of its net income out as a dividend, then its internal growth rate is reduced to only the growth supported by its retained earnings. This reasoning suggests a more general formula for the internal growth rate:

$$\text{Internal Growth Rate} = \left(\frac{\text{Net Income}}{\text{Beginning Assets}}\right) \times (1 - \text{Payout Ratio}) = \text{ROA} \times \text{Retention Rate} \qquad (18.4)$$

[...] **Sustainable Growth Rate Formula.** The sustainable growth rate allows for some external financing. It assumes that no new equity will be issued and that the firm's managers want to maintain the same debt-to-equity ratio. Thus, it tells us how fast the firm can grow by reinvesting its retained earnings and issuing only as much new debt as can be supported by those retained earnings. The formula for the sustainable growth rate is:

$$\text{Sustainable Growth Rate} = \left(\frac{\text{Net Income}}{\text{Beginning Equity}}\right) \times (1 - \text{Payout Ratio}) = \text{ROE} \times \text{Retention Rate} \qquad (18.5)$$

What is asked. (a) The fastest growth the firm can finance with retained earnings alone; (b) the fastest growth with retained earnings plus the debt that keeps D/E constant; (c) (b) again when 40% of income is paid out.

Approach. Berk's equations (18.4)–(18.5): internal growth = ROA × retention, sustainable growth = ROE × retention, both on *beginning-of-period* assets and equity. With retention b, equity grows by $b \cdot NI$; assets can grow at that rate relative to assets (no external money) or relative to equity (debt grows in step).

Notation.

- $NI = 50{,}000$; $A_0 = 400{,}000$, $E_0 = 250{,}000$: beginning assets and equity.
- $b = 1 -$ payout: retention (plowback) ratio; ROA $= NI/A_0$, ROE $= NI/E_0$.

Step 1 — (a) Internal growth. With payout 0%, $b = 1$:

$$g_{int} = \text{ROA} \times b = \frac{50{,}000}{400{,}000} \times 1 = 0.125 = 12.5\%. \tag{8}$$

Answer (a). 12.5%.

Step 2 — (b) Sustainable growth.

$$g_{sus} = \text{ROE} \times b = \frac{50{,}000}{250{,}000} \times 1 = 0.20 = 20\%. \tag{9}$$

Answer (b). 20%.

Step 3 — (c) Payout 40%. Now $b = 1 - 0.40 = 0.60$:

$$g_{sus} = (0.20)(0.60) = 0.12 = 12\%. \tag{10}$$

Answer (c). 12%.

Intuition. Retained earnings are the only internal source of equity. Measured against assets they bound growth without any outside money; measured against equity they bound growth when debt is allowed to grow in proportion. Leverage ($A_0/E_0 = 1.6$) is why (b) exceeds (a): $20\% = 12.5\% \times 1.6$.

Common mistakes.

- Using end-of-year equity or assets with the simple formulas (then the formulas are $xb/(1-xb)$; see Exercise 13).
- Forgetting to multiply by the retention ratio when there is a payout.

Ref: Berk §18.4, Eqs. (18.4)–(18.5), p. 548; Example 18.3, p. 549.

5. Universal Products: finishing a pro forma — Higgins 3.5, p. 104

Statement.

Source: Higgins, Ch. 3, Exercise 3.5, p. 104 (PDF 125) · Answer in book: yes, p. 405

5. Universal Products Co. has almost finished its pro forma financial statements for 2022, as shown next.

Universal Products Co. — Financial Statements, 2021 and Pro Forma 2022 ($ thousands)

INCOME STATEMENT	Actual 2021	Forecast 2022	BALANCE SHEET	Actual 2021	Forecast 2022
Sales	$5,000	$5,500	Current assets	$2,500	$2,750
Cost of goods sold	4,000	4,400	Net fixed assets	2,000	_____
Operating expense	500	550	Total assets	$4,500	
Depreciation expense	250	260			
Interest expense	50	50	Current liabilities	$1,050	$1,155
Earnings before tax	200	240	Long-term debt	500	500
Tax	70	84	Owners' equity	2,950	_____

INCOME STATEMENT	Actual 2021	Forecast 2022	BALANCE SHEET	Actual 2021	Forecast 2022
Net income	\$130	\$156	Total liabilities & equity	\$4,500	

a. Assume that Universal plans to purchase \$500,000 in fixed assets during 2022 and to dispose of no fixed assets during 2022. What would be its forecast for net fixed assets in 2022?

b. Assume that Universal plans to have a dividend payout ratio of 50 percent in 2022 and will neither sell nor repurchase equity during 2022. What would be its forecast for owners' equity in 2022?

c. Given the assumptions in questions (a) and (b), what is Universal's projected external funding required for 2022?

What is asked. (a) Net fixed assets at the end of 2022 given \$500 thousand of purchases. (b) Owners' equity with a 50% payout and no equity issues. (c) The resulting external funding required.

Approach. Two roll-forward identities fill the blanks: fixed assets evolve by capex less depreciation; equity by retained earnings. EFN is then the plug of the balance sheet (Exercise 2).

Notation.

- NFA: net fixed assets; $CapEx = 500$; $Dep_{2022} = 260$ (\$ thousands).
- $NI_{2022} = 156$; payout $= 50\%$; E: owners' equity; CA, CL, LTD as in the table.

Step 1 — (a) Fixed assets. Net fixed assets grow by purchases and fall by depreciation:

$$NFA_{2022} = NFA_{2021} + CapEx - Dep_{2022} = 2{,}000 + 500 - 260 = 2{,}240. \tag{11}$$

Answer (a). \$2,240 thousand.

Step 2 — (b) Equity. Equity grows by retained income only:

$$E_{2022} = E_{2021} + NI_{2022}(1 - 0.5) = 2{,}950 + 156(0.5) = 2{,}950 + 78 = 3{,}028. \tag{12}$$

Answer (b). \$3,028 thousand.

Step 3 — (c) The plug. Totals and the difference:

$$\begin{aligned} TA_{2022} &= CA + NFA = 2{,}750 + 2{,}240 = 4{,}990, \\ (L+E)_{2022} &= CL + LTD + E = 1{,}155 + 500 + 3{,}028 = 4{,}683, \\ EFN_{2022} &= 4{,}990 - 4{,}683 = 307. \end{aligned} \tag{13}$$

Answer (c). External funding required = \$307 thousand.

Intuition. Assets grow by $4{,}990 - 4{,}500 = 490$, but spontaneous liabilities supply only $1{,}155 - 1{,}050 = 105$ and retained earnings 78; the remaining $490 - 105 - 78 = 307$ must come from outside.

Common mistakes.

- Adding all of net income to equity (ignoring the 50% dividend).
- Forgetting depreciation in the fixed-asset roll-forward, or subtracting 2021's (250) instead of 2022's (260).

Book answer. Matches the book (p. 405): $NFA = 2{,}240$, equity 3,028, EFN 307.

Ref: Higgins Ch. 3, "Pro Forma Statements", pp. 80–88.

6. Net new financing and how to split it — Berk 18.2 and 18.3, p. 558

Statement.

Source: Berk, Ch. 18, Exercises 18.2 and 18.3, p. 558 (PDF 589) · Answer in book: no

2. For the next fiscal year, you forecast net income of $50,000 and ending assets of $500,000. Your firm's payout ratio is 10%. Your beginning stockholders' equity is $300,000 and your beginning total liabilities are $120,000. Your non-debt liabilities such as accounts payable are forecasted to increase by $10,000. What is your net new financing needed for next year?
3. Assume your beginning debt in Problem 2 is $100,000. What amount of equity and what amount of debt would you need to issue to cover the net new financing in order to keep your debt-equity ratio constant?

Context needed to solve it *From Berk §18.2, "Pro Forma Balance Sheet", p. 539 (PDF 570):*

The imbalance indicates that we will need $8396 in *net new financing* to fund our growth. **Net new financing** is the amount of additional external financing we will need to secure to pay for the planned increase in assets. It can be computed as:

$$\text{Net New Financing} = \text{Projected Assets} - \text{Projected Liabilities and Equity}$$

What is asked. (18.2) The external financing needed next year given ending assets, net income, payout and the growth of non-debt liabilities. (18.3) With beginning debt of $100,000, how much of that financing must be debt and how much equity to keep the debt-equity ratio unchanged.

Approach. Berk's identity: Net New Financing = Projected Assets − Projected Liabilities and Equity, where projected liabilities and equity include only the *spontaneous* sources (payables) and retained earnings. Part 18.3 is a financing-policy choice: the total is fixed by 18.2; the policy splits it.

Notation.

- $TA_1 = 500{,}000$; $NI = 50{,}000$; payout 10%; $E_0 = 300{,}000$; $L_0 = 120{,}000$ (of which $D_0 = 100{,}000$ debt in 18.3).
- NNF: net new financing; ΔD, ΔE^{new}: new debt and new equity issued.

Step 1 — (18.2) Projected sources. Retained earnings are $50{,}000(1 - 0.10) = 45{,}000$, so

$$E_1 = 300{,}000 + 45{,}000 = 345{,}000, \qquad L_1 = 120{,}000 + 10{,}000 = 130{,}000,$$
$$NNF = TA_1 - (L_1 + E_1) = 500{,}000 - (130{,}000 + 345{,}000) = 500{,}000 - 475{,}000 = 25{,}000. \quad (14)$$

Answer (18.2). Net new financing = $25,000.

Step 2 — (18.3) Target capital structure. Non-debt liabilities are $120{,}000 - 100{,}000 = 20{,}000$ at the start and $20{,}000 + 10{,}000 = 30{,}000$ at the end. Debt plus equity must therefore fund $500{,}000 - 30{,}000 = 470{,}000$ of assets, in the initial proportion $D/E = 100{,}000/300{,}000 = 1/3$, i.e. $D = \frac{1}{4}$ and $E = \frac{3}{4}$ of the total:

$$E_1 = \tfrac{3}{4}(470{,}000) = 352{,}500, \qquad D_1 = \tfrac{1}{4}(470{,}000) = 117{,}500. \quad (15)$$

Step 3 — (18.3) Issues. Equity already reaches 345,000 through retention; debt starts at 100,000:

$$\Delta E^{new} = 352{,}500 - 345{,}000 = 7{,}500, \qquad \Delta D = 117{,}500 - 100{,}000 = 17{,}500,$$

and $7{,}500 + 17{,}500 = 25{,}000 = NNF$.

Answer (18.3). Issue $7,500 of equity and $17,500 of debt.

Intuition. Net new financing is fixed by the operating plan; how it is raised is a separate capital structure decision. Keeping D/E constant forces some equity issuance here because retained earnings (45,000) are too small relative to the asset growth.

Common mistakes.

- Treating payables as debt when computing D/E; they are spontaneous operating liabilities.
- Applying $D/E = 1/3$ to the \$25,000 alone; the ratio must hold for the *stocks*, and retained earnings already added equity.

Ref: Berk §18.2, "Pro Forma Balance Sheet" and "Making the Balance Sheet Balance", pp. 539–541.

7. Companhia João Herdeiro: percent-of-sales balance sheet — Ross 4.9–4.11, pp. 118–119

Statement.

Source: Ross, Cap. 4, Exercises 4.9–4.11, pp. 118–119 (PDF 142–143) · Answer in book: no (selected answers in the web-only Apêndice C)

9. **Cálculo dos lucros retidos a partir da demonstração financeira projetada** [OA1] Considere a seguinte demonstração de resultados da Companhia João Herdeiro:

COMPANHIA JOÃO HERDEIRO — Demonstração de resultados

Vendas		\$38.000
Custos		18.400
Lucro tributável		\$19.600
Impostos (34%)		6.664
Lucro líquido		\$12.936
Dividendos	\$5.200	
Acréscimo aos lucros retidos	7.736	

A taxa de crescimento projetada para as vendas é de 20%. Prepare a demonstração de resultados projetada assumindo que os custos variam com as vendas e a taxa de distribuição de dividendos é constante. Qual é a previsão de acréscimo aos lucros retidos?

10. **Aplicação da porcentagem de vendas** [OA1] A seguir está o balanço patrimonial da Companhia João Herdeiro. Com base nestas informações e na demonstração de resultados do problema anterior, forneça as informações que faltam usando a abordagem da porcentagem de vendas. Assuma que as contas a pagar variam com as vendas, mas não os empréstimos a pagar. Coloque "n/a" onde for necessário.

COMPANHIA JOÃO HERDEIRO — Balanço patrimonial

Ativo	\$	Porcentagem de vendas	Passivo e patrimônio líquido	\$	Porcentagem de vendas
Ativo circulante			*Passivo circulante*		
Caixa	\$ 3.050	—	Contas a pagar	\$ 1.300	—
Contas a receber	6.900	—	Empréstimos	6.800	—
Estoque	7.600	—	Total	\$ 8.100	—
Total	\$17.550	—	Passivo não circulante	\$25.000	—
Ativo imobilizado			*Patrimônio líquido*		—
Instalações e equipamentos	\$34.500	—	Capital social e ágio recebido na emissão de ações	\$15.000	—

Ativo	$	Porcentagem de vendas	Passivo e patrimônio líquido	$	Porcentagem de vendas
			Lucros retidos	3.950	—
			Total	$18.950	—
Total do ativo	$52.050	—	Total do passivo e do patrimônio líquido	$52.050	—

11. **NAF e vendas** [OA2] Com as duas questões anteriores, prepare um balanço patrimonial projetado mostrando a NAF. Assuma um aumento de 15% nas vendas, nenhuma dívida externa nova ou venda de novas ações e uma taxa de distribuição de dividendos constante.

What is asked. (4.9) The pro forma income statement at 20% sales growth with constant payout, and the addition to retained earnings. (4.10) Each balance sheet item as a percent of sales, "n/a" where it does not vary with sales. (4.11) The pro forma balance sheet at 15% growth with no new debt or equity, and the NAF.

Approach. Percent of sales: costs, assets and payables scale with sales; notes payable, long-term debt and paid-in capital stay; retained earnings grow by the retained part of the new net income. NAF is the plug.

Notation.

- $S_0 = 38{,}000$, $C_0 = 18{,}400$; tax rate 34%; $Div_0 = 5{,}200$, $NI_0 = 12{,}936$.
- $b = 7{,}736/12{,}936 = 0.598021$: retention ratio; $1 - b = 0.401979$: payout.

Step 1 — (4.9) Income statement at $g = 20\%$.

$$S_1 = 38{,}000(1.2) = 45{,}600, \quad C_1 = 18{,}400(1.2) = 22{,}080, \quad EBT_1 = 45{,}600 - 22{,}080 = 23{,}520,$$
$$T_1 = 0.34(23{,}520) = 7{,}996.8, \qquad NI_1 = 23{,}520 - 7{,}996.8 = 15{,}523.2. \tag{16}$$

Since all lines scale with sales, $NI_1 = 1.2(12{,}936) = 15{,}523.2$ too. With a constant payout, $Div_1 = 15{,}523.2(0.401979) = 6{,}240$ and the addition to retained earnings is $15{,}523.2(0.598021) = 9{,}283.2$ $(= 1.2 \times 7{,}736)$.

Answer (4.9). Addition to retained earnings = $9,283.20 (dividends $6,240).

Step 2 — (4.10) Percent of sales. Divide each item by $S_0 = 38{,}000$:

Ativo	$	% sales	Passivo e PL	$	% sales
Caixa	3,050	8.03	Contas a pagar	1,300	3.42
Contas a receber	6,900	18.16	Empréstimos	6,800	n/a
Estoque	7,600	20.00	Total circulante	8,100	n/a
Total circulante	17,550	46.18	Passivo não circulante	25,000	n/a
Instalações e equip.	34,500	90.79	Capital social	15,000	n/a
			Lucros retidos	3,950	n/a
			Total PL	18,950	n/a
Total do ativo	52,050	136.97	Total	52,050	n/a

The current-liabilities total is n/a because only part of it (payables) varies with sales.

Answer (4.10). See the table: all assets and payables scale (total assets 136.97% of sales); loans, long-term debt and equity are n/a.

Step 3 — (4.11) Income at $g = 15\%$. $S_1 = 43{,}700$, $C_1 = 21{,}160$, $EBT_1 = 22{,}540$, $T_1 = 0.34(22{,}540) = 7{,}663.6$, $NI_1 = 22{,}540 - 7{,}663.6 = 14{,}876.4$; dividends $14{,}876.4(0.401979) = 5{,}980$; addition to retained earnings $14{,}876.4(0.598021) = 8{,}896.4$.

Step 4 — (4.11) Pro forma balance sheet. Scaling assets and payables by 1.15 and rolling retained earnings forward $(3{,}950 + 8{,}896.4 = 12{,}846.4)$:

Ativo	$	Passivo e PL	$
Caixa	3,507.5	Contas a pagar	1,495.0
Contas a receber	7,935.0	Empréstimos	6,800.0
Estoque	8,740.0	Total circulante	8,295.0
Total circulante	20,182.5	Passivo não circulante	25,000.0
Instalações e equip.	39,675.0	Capital social	15,000.0
		Lucros retidos	12,846.4
		Total PL	27,846.4
Total do ativo	59,857.5	Total	61,141.4

$$\text{NAF} = 59{,}857.5 - (8{,}295 + 25{,}000 + 27{,}846.4) = 59{,}857.5 - 61{,}141.4 = -1{,}283.9. \tag{17}$$

Answer (4.11). NAF = −$1,283.90: at 15% growth the firm generates a surplus.

Intuition. Asset needs rise by $0.15 \times 52{,}050 = 7{,}807.5$; payables supply 195 and retained earnings 8,896.4, more than enough. A very profitable firm (net margin 34%) can finance moderate growth internally; the surplus would turn into a deficit at a higher growth rate (Exercise 14 shows how to find the switch point).

Common mistakes.

- Using the 20% income statement of 4.9 in 4.11; the growth rate changes to 15%.
- Scaling loans (*empréstimos*) with sales: they are a financing decision.

Ref: Ross §4.3, pp. 99–104.

8. Global Corp.: two-pass pro forma — Berk 18.8–18.11, p. 560

Statement.

Source: Berk, Ch. 18, Exercises 18.8–18.11, p. 560 (PDF 591) · Answer in book: no

For Problems 8–11, use the following income statement and balance sheet for Global Corp.:

Figures in $ millions

Income statement		Balance sheet	
Net sales	186.7	*Assets*	
Costs Except Depreciation	−175.1	Cash	23.2
EBITDA	11.6	Accounts Receivable	18.5
Depreciation and Amortization	−1.2	Inventories	15.3
EBIT	10.4	Total Current Assets	57
Interest Income (expense)	−7.7		
Pretax Income	2.7	Net Property, Plant, and Equipment	113.1
Taxes	−0.7	Total Assets	170.1
Net Income	2.0		
		Liabilities and Equity	
		Accounts Payable	34.7
		Long-Term Debt	113.2
		Total Liabilities	147.9
		Total Stockholders' Equity	22.2
		Total Liabilities and Equity	170.1

8. Global expects sales to grow by 8% next year. Using the percent of sales method, forecast:

a. Costs except depreciation

b. Depreciation

c. Net income

d. Cash

e. Accounts receivable

f. Inventory

g. Property, plant, and equipment

h. Accounts payable

9. Assume that Global pays out 50% of its net income. Use the percent of sales method to forecast stockholders' equity.

10. What is the amount of net new financing needed for Global?

11. If Global decides that it will limit its net new financing to no more than $9 million, how will this affect its payout policy?

Context needed to solve it *The method these problems apply, from Berk §18.2, pp. 538–540 (PDF 569–571). Text, p. 538:* "In addition to the sales forecast, we require three other details to prepare the pro forma income statement: costs excluding depreciation in 2010 as a percent of sales, depreciation as a percent of sales, and the tax rate." *p. 539:* "The one final assumption we need to make is about our interest expense. We assume for now that it will remain the same as in 2010 because we will determine if our debt needs will change as part of the forecasting process." *p. 540:* "[. . .] our initial assumption is that debt will remain the same [. . .]"

Table 18.2, "KMS Designs' Pro Forma Income Statement for 2011", p. 538 (PDF 569):

Income Statement ($000s)	2010	2011	Calculation
Sales	74,889	88,369	74,889 × 1.18
Costs Except Depreciation	−58,413	−68,928	78% of Sales
EBITDA	16,476	19,441	Lines 3 + 4
Depreciation	−5,492	−6,480	7.333% of Sales
EBIT	10,984	12,961	Lines 5 + 6
Interest Expense (net)	−306	−306	Remains the same
Pretax Income	10,678	12,655	Lines 7 + 8
Income Tax (35%)	−3,737	−4,429	35% of Line 9
Net Income	6,941	8,226	Lines 9 + 10

Table 18.3, "First-Pass Pro Forma Balance Sheet for 2011", p. 540 (PDF 571):

Balance Sheet ($000s)	2010	2011	Calculation
Cash and Cash Equivalents	11,982	14,139	16% of Sales
Accounts Receivable	14,229	16,790	19% of Sales
Inventories	14,978	17,674	20% of Sales
Total Current Assets	41,189	48,603	Lines 4 + 5 + 6
Property, Plant, and Equipment	49,427	58,324	66% of Sales
Total Assets	90,616	106,927	Lines 7 + 8
Accounts Payable	11,982	14,139	16% of Sales
Debt	4,500	4,500	Remains the same
Total Liabilities	16,482	18,639	Lines 11 + 12
Stockholders' Equity	74,134	79,892	74,134 + 70% of 8,226
Total Liabilities and Equity	90,616	98,531	Lines 13 + 14
Net New Financing		8,396	Line 9 − Line 15

Definition, p. 539 (PDF 570): Net New Financing = Projected Assets − Projected Liabilities and Equity.

What is asked. (18.8) Next year's costs, depreciation, net income and the operating balance sheet items at 8% sales growth. (18.9) Equity with a 50% payout. (18.10) Net new financing. (18.11) The payout that keeps net new financing at or below $9 million.

Approach. Berk's KMS template (Tables 18.2–18.3): costs and depreciation as a percent of sales, interest expense unchanged, tax at the current rate; assets and payables as a percent of sales; debt

unchanged in the first pass; equity rolls forward by retained earnings; net new financing is the gap. Part 18.11 inverts the plug: fix NNF, solve for retention.

Notation.

- $g = 8\%$; amounts in \$ millions; $\tau = 0.7/2.7 = 0.259259$: Global's tax rate (taxes/pretax income).
- p: payout; NNF: net new financing.

Step 1 — (18.8) Income statement. Every operating line scales by 1.08; interest stays at 7.7:

$$\begin{aligned} S_1 &= 186.7(1.08) = 201.636, \quad C_1 = 175.1(1.08) = 189.108, \quad Dep_1 = 1.2(1.08) = 1.296, \\ EBIT_1 &= 201.636 - 189.108 - 1.296 = 11.232, \qquad EBT_1 = 11.232 - 7.7 = 3.532, \\ T_1 &= 0.259259(3.532) = 0.915704, \qquad NI_1 = 3.532 - 0.915704 = 2.616296. \end{aligned} \tag{18}$$

(Depreciation is $1.2/186.7 = 0.6427\%$ of sales; scaling by 1.08 is the same thing.)

Step 2 — (18.8) Balance sheet items. Each scales by 1.08: cash $23.2(1.08) = 25.056$; receivables $18.5(1.08) = 19.98$; inventory $15.3(1.08) = 16.524$ (current assets 61.56); PP&E $113.1(1.08) = 122.148$; total assets 183.708; payables $34.7(1.08) = 37.476$.

Answer (18.8). (a) 189.108; (b) 1.296; (c) 2.616; (d) 25.056; (e) 19.98; (f) 16.524; (g) 122.148; (h) 37.476 (\$ millions).

Step 3 — (18.9) Equity. Retained earnings are $0.5(2.616296) = 1.308148$:

$$E_1 = 22.2 + 1.308148 = 23.508148. \tag{19}$$

Answer (18.9). Stockholders' equity $\approx$ \$23.51 million.

Step 4 — (18.10) Net new financing. Debt stays at 113.2 in the first pass:

$$\begin{aligned} (L+E)_1 &= 37.476 + 113.2 + 23.508148 = 174.184148, \\ NNF &= 183.708 - 174.184148 = 9.523852. \end{aligned} \tag{20}$$

Answer (18.10). Net new financing $\approx$ \$9.52 million.

Step 5 — (18.11) Payout cap. Write NNF as a function of the payout: $NNF(p) = 183.708 - 37.476 - 113.2 - 22.2 - (1-p)NI_1 = 10.832 - (1-p)(2.616296)$. Requiring $NNF \le 9$:

$$\begin{aligned} (1-p)(2.616296) &\ge 10.832 - 9 = 1.832, \\ 1 - p &\ge \frac{1.832}{2.616296} = 0.700227, \\ p &\le 0.299773. \end{aligned} \tag{21}$$

Dividends fall from 1.308 to at most $2.616296 - 1.832 = 0.784$ million, i.e. by the 0.524 excess.

Answer (18.11). Global must cut its payout from 50% to at most about 30% (dividends $\le$ \$0.78 million).

Intuition. Global is highly levered with a thin margin (net income \$2.0 million on \$186.7 million of sales), so 8% growth in a \$170 million asset base creates a financing need (13.6 of new assets) that retained earnings barely dent. The payout is the only internal lever left; cutting it from 50% to 30% saves just \$0.52 million.

Common mistakes.

- Growing interest expense with sales; the method holds it fixed until the financing is decided.
- Using a 35% tax rate (the KMS rate) instead of Global's own 0.7/2.7.
- Solving 18.11 by scaling dividends instead of solving the inequality for the payout.

Ref: Berk §18.2, Tables 18.1–18.4 and Example 18.2, pp. 537–541.

9. KMS growth vs. its SGR; new debt; D/E at the internal rate — Berk 18.17–18.19, p. 561

Statement.

Source: Berk, Ch. 18, Exercises 18.17–18.19, p. 561 (PDF 592) · Answer in book: no

17. Did KMS's expansion plan call for it to grow slower or faster than its sustainable growth rate?
18. Your firm has an ROE of 12%, a payout ratio of 25%, \$600,000 of stockholders' equity, and \$400,000 of debt. If you grow at your sustainable growth rate this year, how much additional debt will you need to issue?
19. IZAX, Co. had the following items on its balance sheet at the beginning of the year:

Assets		Liabilities and Equity	
Cash	50,000	Debt	100,000
PPE	350,000	Equity	300,000

Its net income this year is \$20,000 and it pays dividends of \$5,000. If its assets grew at its internal growth rate, what is its new D/E ratio?

Context needed to solve it *Formulas (18.4) and (18.5), Berk §18.4, p. 548 (PDF 579):* Internal Growth Rate = (Net Income/Beginning Assets) × (1 − Payout Ratio) = ROA × Retention Rate; Sustainable Growth Rate = (Net Income/Beginning Equity) × (1 − Payout Ratio) = ROE × Retention Rate. "[. . .] the firm's *sustainable growth rate*—the maximum growth rate the firm can sustain without issuing new equity or increasing its debt-to-equity ratio."

KMS's expansion plan (for Problem 17). Sales and net income from Table 18.8, "Pro Forma Income Statement for KMS Expansion", p. 545 (PDF 576); equity and dividends from Table 18.17, "Pro Forma Balance Sheet for KMS, 2010–2015", p. 562 (PDF 593). \$000s:

Year	2010	2011	2012	2013	2014	2015
Sales (Table 18.8)	74,889	88,369	103,247	119,793	138,167	158,546
Net Income (Table 18.8)	6,941	7,600	8,807	11,141	13,739	16,627
Starting Stockholders' Equity (Table 18.17)	69,275	74,134	76,948	83,252	90,156	97,729
Net Income (Table 18.17)	6,940	7,600	8,807	11,141	13,739	16,627
Dividends (Table 18.17)	−2,082	−4,786	−2,503	−4,237	−6,167	−8,313
Stockholders' Equity (Table 18.17)	74,134	76,948	83,252	90,156	97,729	106,042
Debt (Table 18.17)	4,500	24,500	24,500	24,500	24,500	24,500

From Berk §18.3, p. 544 (PDF 575): KMS finances the \$20 million of new equipment in 2011 with ten-year bonds; "KMS distributes all cash in excess of the minimum required cash as dividends" (p. 545).

What is asked. (18.17) Whether the KMS plan grows sales faster or slower than its sustainable growth rate (SGR). (18.18) How much debt a firm must issue if it grows exactly at its SGR. (18.19) IZAX's new D/E if its assets grow at its internal growth rate.

Approach. SGR = ROE $\times b$ (beginning equity), Eq. (18.5): the growth that leaves D/E constant with no new equity. Comparing planned growth with SGR tells us whether leverage, payout or equity issues must change. At the internal rate no external money is raised, so debt is frozen and D/E falls.

Notation.

- $g_t = S_t/S_{t-1} - 1$: KMS sales growth; $\text{SGR}_t = (NI_t - Div_t)/E_{t-1}$; $\text{ROE}_t = NI_t/E_{t-1}$.
- 18.18: ROE = 12%, payout 25%, $E_0 = 600{,}000$, $D_0 = 400{,}000$.
- 18.19: $A_0 = 400{,}000$, $D_0 = 100{,}000$, $E_0 = 300{,}000$, $NI = 20{,}000$, $Div = 5{,}000$.

Step 1 — (18.17) Growth and SGR year by year. Using Tables 18.8 and 18.17 ($000s); e.g. 2011: $g = 88{,}369/74{,}889 - 1 = 18.00\%$ and SGR $= (7{,}600 - 4{,}786)/74{,}134 = 2{,}814/74{,}134 = 3.80\%$.

	2011	2012	2013	2014	2015
Sales growth g_t (%)	18.00	16.84	16.03	15.34	14.75
SGR with the plan's dividends (%)	3.80	8.19	8.29	8.40	8.51
ROE = SGR at full retention (%)	10.25	11.45	13.38	15.24	17.01

Even under the 2010 policy (30% payout, Section 18.2) the SGR was $6{,}941/69{,}275 \times 0.7 = 7.01\%$.

Answer (18.17). Faster: sales grow 15–18% a year against an SGR of 4–9%. KMS covers the gap by raising leverage — book D/E jumps from $4{,}500/74{,}134 = 6.07\%$ to $24{,}500/76{,}948 = 31.84\%$ in 2011 — and, through 2014, growth exceeds even the full-retention ROE.

Step 2 — (18.18) Debt at the SGR.

$$\begin{aligned} &\text{SGR} = 0.12(1 - 0.25) = 0.09, \qquad \Delta E = 600{,}000(0.12)(0.75) = 54{,}000 = 9\% \text{ of } E_0, \\ &\Delta D = 0.09(400{,}000) = 36{,}000. \end{aligned} \tag{22}$$

Check: $(400{,}000 + 36{,}000)/(600{,}000 + 54{,}000) = 436{,}000/654{,}000 = 2/3 = 400{,}000/600{,}000$.

Answer (18.18). Issue $36,000 of new debt.

Step 3 — (18.19) IZAX at the internal rate.

$$\begin{aligned} &\text{ROA} = 20{,}000/400{,}000 = 0.05, \quad b = 15{,}000/20{,}000 = 0.75, \quad g_{int} = (0.05)(0.75) = 0.0375, \\ &A_1 = 400{,}000(1.0375) = 415{,}000, \quad E_1 = 300{,}000 + 15{,}000 = 315{,}000, \quad D_1 = 100{,}000, \\ &D_1/E_1 = 100{,}000/315{,}000 = 0.3175. \end{aligned} \tag{23}$$

Answer (18.19). $D/E = 0.3175$ (down from 0.3333).

Intuition. The SGR is a benchmark, not a target. KMS's planned growth is far above it, which only says that the plan needs outside money (here a $20 million bond); whether that growth is *worth* it is an NPV question (Exercise 15). At the internal rate equity grows and debt does not, so leverage drifts down; at the sustainable rate debt grows with equity and leverage stays put.

Common mistakes.

- Calling growth above the SGR "bad": Berk stresses that only NPV decides value.
- Computing ROE on end-of-year equity with the simple formula.

Ref: Berk §18.4, pp. 547–551; Tables 18.8, 18.17, pp. 545, 562.

10. Under Armour: sustainable growth and its sensitivity — Higgins 4.8, pp. 135–136

Statement.

Source: Higgins, Ch. 4, Exercise 4.8, pp. 135–136 (PDF 156–157) · Answer in book: no

8. Under Armour, Inc., is an American supplier of sportswear and casual apparel. Following are selected financial data for the company for the period 2009–2013.

	2009	2010	2011	2012	2013
Profit margin (%)	5.5	6.4	6.6	7.0	7.0
Retention ratio (%)	100.0	100.0	100.0	100.0	100.0
Asset turnover (X)	1.6	1.6	1.6	1.6	1.5
Financial leverage (X)	1.7	1.7	1.9	1.8	1.9
Growth rate in sales (%)	18.1	24.2	38.4	24.6	27.1

Source: Data from Under Armour 2009–2013 annual reports

a. Calculate Under Armour's annual sustainable growth rate for the years 2009 through 2013.

b. Did Under Armour face a growth management challenge during this period? Please explain briefly.

c. How did Under Armour cope with this challenge?

d. Calculate Under Armour's sustainable growth rate in 2013 assuming a profit margin of 8.0%. Return the profit margin to its original value and calculate the sustainable growth rate in 2013 assuming a financial leverage of 2.40 times. Calculate the sustainable growth rate in 2013 assuming both of these changes occur.

Context needed to solve it *From Higgins Ch. 4, "Sustainable Growth", pp. 113–114 (PDF 134–135):*

A company's sustainable growth rate therefore is nothing more than its growth rate in equity. Letting g^* represent the sustainable growth rate,

$$g^* = \frac{\text{Change in equity}}{\text{Equity}_{bop}}$$

where bop denotes beginning-of-period equity. Because the firm will not be selling any new shares by assumption, the only source of new equity will be from retained profits, so we can rewrite this expression as

$$g^* = \frac{R \times \text{Earnings}}{\text{Equity}_{bop}}$$

where R is the firm's "retention rate." [...] The ratio "Earnings/Equity" in this expression should look familiar; it is the firm's return on equity, or ROE. Thus, $g^* = R \times \text{ROE}_{bop}$. Finally, recalling the levers of performance discussed in Chapter 2, we can rewrite this expression yet again as

$$g^* = PRA\hat{T}$$

where P, A, and $\hat{T}$ are our old friends from Chapter 2, the levers of performance, where P is the profit margin, A is the asset turnover ratio, and $\hat{T}$ is the assets-to-equity ratio. The assets-to-equity ratio wears a hat here as a reminder that it is assets divided by beginning-of-period equity instead of end-of-period equity as defined in Chapter 2. [...] If a company increases sales at any rate other than g^*, one or more of the ratios must change. This means that when a company grows at a rate in excess of its sustainable growth rate, it had better improve operations (represented by an increase in the profit margin or the asset turnover ratio) or prepare to alter its financial policies (represented by increasing its retention rate or its financial leverage).

What is asked. (a) The sustainable growth rate each year 2009–2013. (b) Whether actual growth outran it. (c) How the company coped. (d) The 2013 SGR with an 8% margin, with leverage of 2.40, and with both.

Approach. Higgins' $g^* = PRA\hat{T}$: profit margin × retention × asset turnover × assets/beginning equity. A firm growing faster than g^* must improve operations (P, A) or change financial policy (R, $\hat{T}$) or issue equity. Part (d) is a one-at-a-time sensitivity analysis.

Notation.

- P (%), $R = 1$, A (times), $\hat{T}$ (times): the four ratios in the table; g: actual sales growth.

Step 1 — (a) The product of four ratios. With $R = 1$ in every year, $g^* = P \cdot A \cdot \hat{T}$:

$$\begin{aligned} g^*_{2009} &= 5.5(1.6)(1.7) = 14.96\%, & g^*_{2010} &= 6.4(1.6)(1.7) = 17.41\%, \\ g^*_{2011} &= 6.6(1.6)(1.9) = 20.06\%, & g^*_{2012} &= 7.0(1.6)(1.8) = 20.16\%, \\ g^*_{2013} &= 7.0(1.5)(1.9) = 19.95\%. \end{aligned} \tag{24}$$

Answer (a). 14.96%, 17.41%, 20.06%, 20.16%, 19.95%.

Step 2 — (b) Compare with actual growth. Actual growth was 18.1, 24.2, 38.4, 24.6 and 27.1%: above g^* every year, by up to $38.4 - 20.06 = 18.34$ points in 2011.

Answer (b). Yes: a too-much-growth challenge — growing faster than equity could be built internally.

Step 3 — (c) Which ratios moved. Retention was already 100%. The margin rose (5.5% → 7.0%) and leverage rose ($\hat{T}$: 1.7 → 1.9); turnover slipped (1.6 → 1.5), which worked the other way. Because each year's g^* uses that year's ratios, growth above g^* in a year must have been met by ratios rising further (or new equity) during that year.

Answer (c). By raising profitability and, above all, financial leverage (more debt per dollar of beginning equity), with no dividends; operating efficiency (turnover) did not help.

Step 4 — (d) Sensitivity of 2013 g^*.

$$\begin{aligned} P = 8.0\%: &\quad g^* = 8.0(1.5)(1.9) = 22.8\%, \\ \hat{T} = 2.40: &\quad g^* = 7.0(1.5)(2.4) = 25.2\%, \\ \text{both}: &\quad g^* = 8.0(1.5)(2.4) = 28.8\%. \end{aligned} \tag{25}$$

Answer (d). 22.8%, 25.2% and 28.8%; only the combination exceeds 2013's actual 27.1%.

Intuition. g^* is linear in each lever, so the levers multiply: a 14% rise in the margin and a 26% rise in leverage together lift g^* by 44%. Leverage is the easiest lever to pull and the riskiest to rely on.

Common mistakes.

- Adding the percentage effects in (d) instead of multiplying the ratios.
- Using end-of-period assets/equity; Higgins' $\hat{T}$ uses beginning equity.

Ref: Higgins Ch. 4, "Sustainable Growth" and "What If Actual Growth Exceeds Sustainable Growth?", pp. 112–123.

11. Turismo da Onça: NAF, capacity, constant D/E — Ross 4.25–4.27, p. 121

Statement.

Source: Ross, Cap. 4, Exercises 4.25–4.27, p. 121 (PDF 145) · Answer in book: no (selected answers in the web-only Apêndice C)

25. **Cálculo da NAF** [OA2] A seguir estão as demonstrações contábeis mais recentes da Turismo da Onça S/A. As vendas de 2009 têm crescimento projetado de 20%. As despesas com juros, a alíquota tributária e a taxa de distribuição de dividendos permanecerão constantes. Os custos, as outras despesas, o ativo circulante e as contas a pagar aumentam na proporção das vendas. Se a empresa opera à capacidade máxima e não foram emitidas ações ou assumidas novas dívidas, qual é o aporte financeiro necessário para suportar a taxa de crescimento de 20% nas vendas?

TURISMO DA ONÇA S/A — Demonstração de resultados de 2008

Vendas		$929.000
Custos		723.000
Outras despesas		19.000
Lucros antes de juros e impostos		$187.000
Juros pagos		14.000
Lucro tributável		$173.000
Impostos		60.550
Lucro líquido		$112.450
Dividendos	$33.735	
Acréscimo aos lucros retidos	78.715	

TURISMO DA ONÇA S/A — Balanço patrimonial de 31 de Dezembro de 2008

Ativo		Passivo e patrimônio líquido	
Ativo circulante		*Passivo circulante*	
Caixa	$ 25.300	Contas a pagar	$ 68.000
Contas a receber	40.700	Empréstimos a pagar	17.000
Estoque	86.900	Total	$ 85.000
Total	$152.900	Passivo não circulante	$158.000
Ativo imobilizado		*Patrimônio líquido*	
Instalações e equipamentos	413.000	Capital social e ágio	$140.000
		Lucros retidos	182.900
		Total	$322.900
Total do ativo	$565.900	Total do passivo e do patrimônio líquido	$565.900

26. **Uso da capacidade e crescimento** [OA2] No problema anterior, suponha que a empresa tenha operado a apenas 80% da capacidade em 2008. Qual é a NAF agora?
27. **Cálculo da NAF** [OA2] No Problema 25, suponha que a empresa queira manter seu índice dívida/capital próprio constante. Qual é a NAF agora?

Context needed to solve it *From Ross §4.3, "Uma situação alternativa" (capacity), p. 104 (PDF 128):*
Se assumirmos que a Mar de Rosas está operando a apenas 70% de sua capacidade, a necessidade de aportes financeiros será bastante diferente. Quando dizemos "70% da capacidade", queremos dizer que o nível atual de vendas é 70% da capacidade total de vendas:

Vendas atuais = $1.000 = 0,70 × Capacidade total de vendas Capacidade total de vendas = $1.000/0,70 = $1.429.

Isso nos mostra que as vendas poderiam aumentar em quase 43% – de $1.000 para $1.429 – antes que fossem necessários novos ativos imobilizados.

What is asked. (4.25) The NAF for 20% growth at full capacity with no new debt or equity, interest, tax rate and payout constant. (4.26) The NAF if the firm ran at 80% of capacity. (4.27) The NAF if the firm keeps its debt-equity ratio constant.

Approach. Percent of sales for costs, other expenses, current assets and payables; fixed assets scale only if there is no idle capacity; interest is *fixed*. In 4.27 the financing policy allows debt to grow with equity, so part of the gap is filled by planned borrowing.

Notation.

- $S_0 = 929{,}000$; $\tau = 60{,}550/173{,}000 = 0.35$; payout $= 33{,}735/112{,}450 = 0.30$; interest 14,000.
- "Total debt" (Ross) = all liabilities: $D_0 = 85{,}000 + 158{,}000 = 243{,}000$; $E_0 = 322{,}900$.

Step 1 — Income statement at $g = 20\%$.

$$S_1 = 1{,}114{,}800, \quad \text{Custos} = 867{,}600, \quad \text{Outras} = 22{,}800, \quad EBIT = 224{,}400,$$
$$EBT = 224{,}400 - 14{,}000 = 210{,}400, \quad T = 0.35(210{,}400) = 73{,}640, \quad NI = 136{,}760,$$
$$Div = 0.30(136{,}760) = 41{,}028, \qquad \text{Acréscimo aos lucros retidos} = 95{,}732. \tag{26}$$

Step 2 — (4.25) Balance sheet at full capacity. Current assets 152,900(1.2) = 183,480 (cash 30,360, receivables 48,840, inventory 104,280); fixed assets 413,000(1.2) = 495,600; total 679,080. Payables 81,600; notes 17,000; long-term 158,000; equity 140,000 + 182,900 + 95,732 = 418,632; total 81,600 + 17,000 + 158,000 + 418,632 = 675,232.

$$\text{NAF} = 679{,}080 - 675{,}232 = 3{,}848. \tag{27}$$

Answer (4.25). NAF = \$3,848.

Step 3 — (4.26) Idle capacity. Full-capacity sales are 929,000/0.80 = 1,161,250 > 1,114,800, so no new fixed assets are needed (utilization rises to 96%). Total assets = 183,480 + 413,000 = 596,480:

$$\text{NAF} = 596{,}480 - 675{,}232 = -78{,}752. \tag{28}$$

Answer (4.26). NAF = −\$78,752 (a surplus).

Step 4 — (4.27) Constant total-debt/equity. $D/E = 243{,}000/322{,}900 = 0.752555$. With equity 418,632, permitted total debt is 0.752555(418,632) = 315,043.59; total sources 418,632 + 315,043.59 = 733,675.59:

$$\text{NAF} = 679{,}080 - 733{,}675.59 = -54{,}595.59. \tag{29}$$

Existing and spontaneous liabilities total 81,600 + 17,000 + 158,000 = 256,600, so the policy allows 315,043.59 − 256,600 = 58,443.59 of new borrowing, which more than covers the 3,848 gap.

Answer (4.27). NAF ≈ −\$54,596: the debt the policy allows exceeds the need by that amount.

Intuition. Three different answers from one forecast: the NAF depends as much on assumptions about capacity and financing policy as on the sales forecast. Idle capacity is worth \$82,600 of financing here (495,600 − 413,000).

Common mistakes.

- Growing interest expense with sales (the statement holds it constant).
- In 4.26, cutting fixed assets below the current level; they stay at 413,000.
- In 4.27, applying D/E only to interest-bearing debt; Ross' ratio uses total liabilities (the alternative definition changes the number, so state it).

Ref: Ross §4.3, "Uma situação alternativa", pp. 103–104; §4.4, pp. 106–111.

12. R&E Supplies: extending the spreadsheet, sensitivity and scenario — Higgins 3.9 (3.8), p. 105

Statement.

Source: Higgins, Ch. 3, Exercise 3.9, p. 105 (PDF 126); Part 0 = Exercise 3.8, p. 105 (PDF 126) · Answer in book: yes for 3.9, p. 406; no for 3.8

9. A spreadsheet containing R&E Supplies' 2022 pro forma financial forecast, as shown in Table 3.5, is available for download from McGraw-Hill's *Connect* or your course instructor (see the Preface for more information). Using the spreadsheet information presented next, and the modified equations determined in Problem 8 earlier, extend the forecast for R&E Supplies contained in Table 3.5 through 2023.

R&E Supplies Assumptions for 2023 (\$ thousands)

Growth rate in net sales	30.0%	Tax rate	25.0%
Cost of goods sold/net sales	86.0%	Dividend/earnings after tax	50.0%
Gen., sell., & admin. expenses/net sales	11.0%	Current assets/net sales	29.0%
Long-term debt	\$560	Net fixed assets	\$ 270

Current portion long-term debt	$100	Current liabilities/net sales	14.4%
Interest rate	10.0%		

a. What is R&E's projected external financing required in 2023? How does this number compare to the 2022 projection?

b. Perform a sensitivity analysis on this projection. How does R&E's projected external financing required change if the ratio of cost of goods sold to net sales declines from 86.0 percent to 84.0 percent?

c. Perform a scenario analysis on this projection. How does R&E's projected external financing required change if a severe recession occurs in 2023? Assume net sales decline 5 percent, cost of goods sold rises to 88 percent of net sales due to price cutting, and current assets increase to 35 percent of net sales as management fails to cut purchases promptly in response to declining sales.

Context needed to solve it Part 0 — *Source: Higgins, Ch. 3, Exercise 3.8, p. 105 (PDF 126) · Answer in book: no*

8. Table 3.5 presents a computer spreadsheet for estimating R&E Supplies' external financing required for 2022. The text mentions that with modifications to the equations for equity and net sales, the forecast can easily be extended through 2023. By referencing the appropriate cells, write the modified equations for equity and net sales for 2023.

Table 3.5, "Forecasting with a Spreadsheet: Pro Forma Financial Forecast for R&E Supplies, Inc., December 31, 2022 ($ thousands)", p. 90 (PDF 111). Column A = labels, B = 2021 actual (rows 3–15) or the equations (rows 19–40), C = 2022, D = 2023 (blank):

Row	A	B	C	D
2	Year	2021 Actual	2022	2023
3	Net sales	$20,613		
4	Growth rate in net sales		25.0%	
5	Cost of goods sold/net sales		86.0%	
6	Gen., sell., and admin. expenses/net sales		12.0%	
7	Long-term debt	$ 760	$660	
8	Current portion long-term debt	$ 100	$100	
9	Interest rate		10.0%	
10	Tax rate		25.0%	
11	Dividend/earnings after tax		50.0%	
12	Current assets/net sales		29.0%	
13	Net fixed assets		$280	
14	Current liabilities/net sales		14.5%	
15	Owners' equity	$ 1,730		
16	INCOME STATEMENT			
17		Equations	Forecast	Forecast
18	Year	2022	2022	2023
19	Net sales	= B3 + B3 * C4	$25,766	
20	Cost of goods sold	= C5*C19	22,159	
21	Gross profit	= C19 – C20	3,607	
22	Gen., sell., and admin. exp.	= C6*C19	3,092	
23	Interest expense	= C9*(C7 + C8 + C40)	228	
24	Earnings before tax	= C21 – C22 – C23	287	
25	Tax	= C10*C24	72	
26	Earnings after tax	= C24 – C25	216	
27	Dividends paid	= C11*C26	108	
28	Additions to retained earnings	= C26 – C27	108	
30	BALANCE SHEET			
31	Current assets	= C12*C19	7,472	
32	Net fixed assets	= C13	280	
33	Total assets	= C31 + C32	7,752	
35	Current liabilities	= C14*C19	3,736	
36	Long-term debt	= C7	660	

Row	A	B	C	D
37	Owners' equity	= B15 + C28	1,838	
38	Total liabilities and owners' equity	= C35 + C36 + C37	6,234	
40	EXTERNAL FUNDING REQUIRED	= C33 – C38	$ 1,518	

From Higgins Ch. 3, "Forecasting with Spreadsheets", pp. 89–92 (PDF 110–113):

The first area on the simulated screen is an assumptions box, containing all of the information and assumptions required to construct the forecast. [...] The forecast begins immediately below the assumptions box. The first column, labeled "Equations 2022," is included for explanatory purposes and would not appear on a conventional forecast. Entering the equations shown causes the computer to calculate the quantities appearing in the second column, labeled "Forecast 2022." The third column, labeled "Forecast 2023," is presently blank. [...] The first equation for net sales is = B3 + B3 * C4. This instructs the spreadsheet to get the number in cell B3 and add to it that number times the number in cell C4—in other words, \$20,613 + \$20,613 × 25%. [...] There are only three tricky equations. Interest expense, row 23, is the interest rate times end-of-period long-term debt, including the current portion, plus the forecasted external funding required. As discussed earlier, the tricky part here is the interdependence between interest expense and external funding required. [...] The equity equation, row 37, is end-of-period equity plus additions to retained earnings; the external funding required equation, row 40, is total assets minus total liabilities and owners' equity.

What is asked. (3.8, Part 0) The 2023 spreadsheet formulas for net sales and equity. (a) The 2023 external financing required (EFR) and its comparison with 2022. (b) EFR if COGS falls to 84% of sales. (c) EFR in a recession: sales −5%, COGS 88%, current assets 35% of sales.

Approach. Extend Table 3.5 one column: 2023 sales grow from 2022 forecast sales; 2023 equity starts from 2022 forecast equity. Interest is charged on long-term debt, its current portion *and* the EFR itself, which makes EFR circular; we solve the circularity in closed form instead of iterating. (b) changes one input (sensitivity); (c) changes several together (scenario).

Notation.

- X: 2023 EFR (\$ thousands); $i = 10\%$; $\tau = 25\%$; payout 50%; $LTD = 560$, $CPLTD = 100$, $NFA = 270$.
- S, GP, SGA, CA, CL: 2023 sales, gross profit, SG&A (11%), current assets, current liabilities (14.4%).
- $S_{22} = 20{,}613(1.25) = 25{,}766.25$ and $E_{22} = 1{,}837.81$: Table 3.5's unrounded 2022 values (the table prints 25,766 and 1,838).

Step 0 — (3.8) Formulas. `D19 = C19 + C19*D4` (2023 sales from 2022 forecast sales) and `D37 = C37 + D28` (2023 equity = 2022 forecast equity + 2023 additions to retained earnings).

Answer (3.8). `D19 = C19 + C19*D4`; `D37 = C37 + D28`.

Step 1 — Closed form for the circular interest. EFR $= TA - CL - LTD - E$, with $E = E_{22} + (1-0.5)(1-\tau)[GP - SGA - i(LTD + CPLTD + X)]$. Let $k = (0.5)(0.75) = 0.375$. Then

$$X = TA - CL - LTD - E_{22} - k[GP - SGA - i(660)] + k\,i\,X,$$

$$X(1 - k\,i) = \underbrace{TA - CL - LTD - E_{22} - k[GP - SGA - 66]}_{K}, \qquad X = \frac{K}{1 - 0.0375} = \frac{K}{0.9625}. \quad (30)$$

Step 2 — (a) Base case. $S = 25{,}766.25(1.30) = 33{,}496.13$; $COGS = 0.86S = 28{,}806.67$; $GP = 4{,}689.46$; $SGA = 0.11S = 3{,}684.57$; $CA = 0.29S = 9{,}713.88$; $TA = 9{,}713.88 + 270 = 9{,}983.88$; $CL = 0.144S = 4{,}823.44$.

$$K = 9{,}983.88 - 4{,}823.44 - 560 - 1{,}837.81 - 0.375(4{,}689.46 - 3{,}684.57 - 66) = 2{,}410.54,$$
$$X = 2{,}410.54/0.9625 = 2{,}504.46. \quad (31)$$

Then interest $= 0.1(660 + 2{,}504.46) = 316.45$; EBT $= 4{,}689.46 - 3{,}684.57 - 316.45 = 688.44$; tax 172.11; EAT 516.33; dividends 258.16; equity $1{,}837.81 + 258.16 = 2{,}095.97$; liabilities and equity $4{,}823.44 + 560 + 2{,}095.97 = 7{,}479.42$; and $9{,}983.88 - 7{,}479.42 = 2{,}504.46 = X$, as it must.

Answer (a). EFR ≈ \$2,504 thousand in 2023, about \$986 thousand more than 2022's \$1,518 thousand.

Step 3 — (b) Sensitivity: COGS 84%. Only GP changes: $GP = 0.16S = 5{,}359.38$, so $K = 2{,}410.54 - 0.375(5{,}359.38 - 4{,}689.46) = 2{,}159.32$ and $X = 2{,}159.32/0.9625 = 2{,}243.45$.

Answer (b). EFR ≈ \$2,243 thousand: −\$261 thousand, or −10.4%.

Step 4 — (c) Recession scenario. $S = 25{,}766.25(0.95) = 24{,}477.94$; $COGS = 0.88S = 21{,}540.59$; $GP = 2{,}937.35$; $SGA = 2{,}692.57$; $CA = 0.35S = 8{,}567.28$; $TA = 8{,}837.28$; $CL = 3{,}524.82$.

$$K = 8{,}837.28 - 3{,}524.82 - 560 - 1{,}837.81 - 0.375(2{,}937.35 - 2{,}692.57 - 66) = 2{,}847.60,$$
$$X = 2{,}847.60/0.9625 = 2{,}958.55. \tag{32}$$

Interest = 361.85, EBT = −117.08, EAT = −87.81, and the spreadsheet's dividend formula `=C11*C26` pays −43.90 (a negative dividend).

Answer (c). EFR ≈ \$2,959 thousand, +18.1% over the base case. If dividends are set to zero when earnings are negative (the realistic policy), EFR is ≈ \$3,006 thousand.

Intuition. The financing need is far more sensitive to working capital than to margins: the recession *lowers* sales but raises EFR because current assets jump to 35% of a smaller sales figure and profits vanish. Sensitivity analysis shows which inputs drive the answer; scenario analysis shows how they move together.

Common mistakes.

- Starting 2023 from 2021 data (sales \$20,613, equity \$1,730) instead of the 2022 forecast.
- Ignoring the interest on the EFR itself (understates EFR by about $0.0375X \approx$ \$94 thousand).
- Copying a formula that pays negative dividends without noticing.

Book answer. Matches the book (p. 406): \$2.504 million; \$2.243 million (−10.4%); \$2.959 million (+18.2% in the book; +18.1% here from unrounded values). The book's recession figure uses the spreadsheet's mechanical (negative) dividend.

Ref: Higgins Ch. 3, "Forecasting with Spreadsheets" and "Coping with Uncertainty", pp. 89–98.

13. Deriving NAF and the growth formulas — Ross 4.31–4.33, p. 122

Statement.

Source: Ross, Cap. 4, Exercises 4.31–4.33 (DESAFIO), p. 122 (PDF 146) · Answer in book: no (selected answers in the web-only Apêndice C)

31. **NAF** [OA2] Defina o seguinte:

V = Vendas do ano anterior A = Total do ativo D = Dívida total P = Total do patrimônio líquido g = Projeção de crescimento das vendas ML = Margem de lucro b = Taxa de retenção

Mostre que a NAF pode ser escrita da seguinte forma:

$$\text{NAF} = -\text{ML}(V)b + (A - \text{ML}(V)b) \times g$$

Dica: as necessidades de ativo serão iguais a $A \times g$. O acréscimo aos lucros retidos será igual a $\text{ML}(V)b \times (1 + g)$.

32. **Taxas de crescimento** [OA2] Com base no resultado do Problema 31, mostre que as taxas de crescimento interna e sustentável estão de acordo com aquelas mostradas no capítulo. Dica: para a taxa de crescimento interna, defina NAF igual a zero e calcule g.

33. **Taxa de crescimento sustentável** [OA3] Neste capítulo, discutimos as duas versões da fórmula da taxa de crescimento sustentável. Derive a fórmula ROE × b a partir da fórmula dada no capítulo, onde o ROE se baseia no patrimônio líquido inicial. Da mesma maneira, derive a fórmula ROA × b a partir da fórmula da taxa de crescimento interna.

Context needed to solve it *From Ross §4.4, "Política financeira e crescimento", p. 109 (PDF 133), and Tabela 4.9, p. 111 (PDF 135):*

$$\text{Taxa de crescimento interna} = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b} \qquad [4.2]$$

onde o ROA é o retorno sobre o ativo discutido no Capítulo 3, e b é a taxa de retenção discutida anteriormente neste capítulo. [. . .]

$$\text{Taxa de crescimento sustentável} = \frac{\text{ROE} \times b}{1 - \text{ROE} \times b} \qquad [4.3]$$

Isso é idêntico à taxa de crescimento interna, exceto que é usado o retorno sobre o patrimônio líquido (ROE) em vez do ROA.

Tabela 4.9: ROA = Retorno sobre o ativo = Lucro líquido / Total do ativo; ROE = Retorno sobre o patrimônio líquido = Lucro líquido/Total do patrimônio líquido; b = Taxa de retenção = Acréscimo aos lucros retidos/Lucro líquido. "A taxa de crescimento interna é a taxa máxima de crescimento que pode ser atingida sem qualquer aporte financeiro." "A taxa de crescimento sustentável é a taxa máxima de crescimento que uma empresa pode atingir sem precisar emitir novas ações, enquanto mantém um índice dívida/capital próprio constante."

From Ross, "Uma observação sobre os cálculos da taxa de crescimento sustentável", p. 112 (PDF 136):

Usualmente, a taxa de crescimento sustentável é calculada utilizando apenas o numerador de nossa equação, (ROE × b). Isso causa alguma confusão, que podemos esclarecer aqui. A questão relaciona-se com como o ROE é calculado. lembre-se de que ele é calculado pela divisão do lucro líquido pelo total do patrimônio líquido. Caso o total do patrimônio líquido seja tomado dos números de fim de período do balanço patrimonial (como temos feito até aqui e, em geral, é feito na prática), nossa fórmula está correta. Entretanto, se o total do patrimônio líquido corresponde ao início do período, então a fórmula simplificada será a correta.

What is asked. (4.31) Show NAF $= -ML(V)b + (A - ML(V)b)g$. (4.32) Use it to recover the internal and sustainable growth formulas [4.2]–[4.3]. (4.33) Show that the end-of-period formula $\text{ROE}\,b/(1 - \text{ROE}\,b)$ equals $\text{ROE}_{bop} \times b$, and similarly for ROA.

Approach. The NAF is uses minus internal sources. All assets scale with sales (asset need Ag), the margin is constant, and no liability is spontaneous; retained earnings are next year's income times b. The sustainable case adds the debt that keeps D/P constant.

Notation.

- V: last year's sales; A, D, P: last year's total assets, debt, equity ($A = D + P$).
- g: sales growth; ML: profit margin; b: retention; ROA $= ML \cdot V/A$, ROE $= ML \cdot V/P$ (Ross' definitions, end of the base year).

Step 1 — (4.31) Uses and sources. New assets are Ag; next year's net income is $ML \cdot V(1+g)$, of which b is retained. So

$$\begin{aligned} \text{NAF} &= Ag - ML(V)b(1+g) \\ &= Ag - ML(V)b - ML(V)b\,g \\ &= -ML(V)b + (A - ML(V)b)g. \end{aligned} \qquad (33)$$

Answer (4.31). Shown in (33).

Step 2 — (4.32) Internal growth. Set NAF $= 0$ in (33):

$$(A - ML(V)b)g = ML(V)b \implies g = \frac{ML(V)b}{A - ML(V)b}.$$

Divide numerator and denominator by A and use $\text{ROA} = ML \cdot V/A$:

$$g_{int} = \frac{(ML \cdot V/A)\, b}{1 - (ML \cdot V/A)\, b} = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b}. \tag{34}$$

Step 3 — (4.32) Sustainable growth. Keeping D/P constant lets debt rise by (D/P) per dollar of retained earnings, so the sources are $(1 + D/P)\, ML(V)b(1 + g)$ and $1 + D/P = A/P$. Setting uses equal to sources:

$$Ag = \frac{A}{P} ML(V)b(1+g) \implies g\Big(A - \frac{A}{P} ML(V)b\Big) = \frac{A}{P} ML(V)b,$$
$$g_{sus} = \frac{(ML \cdot V/P)\, b}{1 - (ML \cdot V/P)\, b} = \frac{\text{ROE} \times b}{1 - \text{ROE} \times b} \quad \text{(dividing by } A\text{)}. \tag{35}$$

Illustration (Ross' Hoffman example: $V = A = 500$, $NI = 66$, $P = 250$, $b = 2/3$): $g_{int} = 9.65\%$, $g_{sus} = 21.36\%$.

Answer (4.32). Both chapter formulas [4.2] and [4.3] follow from (33).

Step 4 — (4.33) Beginning- vs. end-of-period ROE. With no new shares, end equity is $P_e = P_b + b\, NI$. Let $\text{ROE}_e = NI/P_e$. Then

$$\frac{\text{ROE}_e\, b}{1 - \text{ROE}_e\, b} = \frac{b\, NI/P_e}{1 - b\, NI/P_e} = \frac{b\, NI}{P_e - b\, NI} = \frac{b\, NI}{P_b} = \text{ROE}_b \times b. \tag{36}$$

The same steps with assets work when assets grow only by retained earnings (no external financing, which is exactly the internal-growth case), $A_e = A_b + b\, NI$: $\text{ROA}_e b/(1 - \text{ROA}_e b) = b\, NI/(A_e - b\, NI) = b\, NI/A_b = \text{ROA}_b \times b$. Ross' numbers: $NI = 20$, $b = 0.6$, $P_b = 80$ give $0.6(20/80) = 15\%$ and $0.6(0.2174)/(1 - 0.6(0.2174)) = 15\%$; with $A_b = 100$, 12%.

Answer (4.33). (36) and its ROA analogue.

Intuition. The "$1 - \text{ROE}\, b$" denominator is just the conversion from end-of-period to beginning-of-period equity: retained earnings are in the end balance but cannot have financed themselves. Use one consistent definition and you get the same growth rate.

Common mistakes.

- Forgetting the $(1 + g)$ on next year's earnings, which drops the $-ML(V)b\, g$ term.
- Mixing definitions: end-of-period ROE in the simple $\text{ROE} \times b$ understates growth ($0.6 \times 0.2174 = 13.0\%$ instead of 15% in Ross' example); beginning-of-period ROE in [4.3] overstates it ($0.15/(1 - 0.15) = 17.6\%$).

Ref: Ross §4.4, pp. 106–113, Tabela 4.9.

14. Turismo da Onça: NAF as a function of growth — Ross 4.28 and 4.29, p. 121

Statement.

Source: Ross, Cap. 4, Exercises 4.28 and 4.29 (DESAFIO), p. 121 (PDF 145); Part 0 = Exercises 4.25 and 4.27, p. 121 (PDF 145) · Answer in book: no (selected answers in the web-only Apêndice C)

28. **NAF e crescimento interno** [OA2, OA3] Refaça o Problema 25 usando agora, além da taxa de crescimento de vendas de 20%, as taxas de 15 e 25%. Ilustre graficamente a relação entre a NAF e a taxa de crescimento. A qual taxa de crescimento a NAF é igual a zero? Por que essa taxa de crescimento interna é diferente daquela encontrada com o uso da equação no texto?
29. **NAF e crescimento interno** [OA2, OA3] Refaça o Problema 27 usando taxas de crescimento de vendas de 30 e 35%, além da de 20%. Ilustre graficamente a relação entre a NAF e a taxa de crescimento. A qual taxa de crescimento a NAF é igual a zero? Por que essa taxa de crescimento sustentável é diferente daquela que encontramos usando a equação no texto?

Context needed to solve it **Part 0** — *Source: Ross, Cap. 4, Exercises 4.25 and 4.27, p. 121 (PDF 145) · Answer in book: no (these are also exercise 11 of this list)*

25. **Cálculo da NAF** [OA2] A seguir estão as demonstrações contábeis mais recentes da Turismo da Onça S/A. As vendas de 2009 têm crescimento projetado de 20%. As despesas com juros, a alíquota tributária e a taxa de distribuição de dividendos permanecerão constantes. Os custos, as outras despesas, o ativo circulante e as contas a pagar aumentam na proporção das vendas. Se a empresa opera à capacidade máxima e não foram emitidas ações ou assumidas novas dívidas, qual é o aporte financeiro necessário para suportar a taxa de crescimento de 20% nas vendas?

TURISMO DA ONÇA S/A — Demonstração de resultados de 2008

Vendas		$929.000
Custos		723.000
Outras despesas		19.000
Lucros antes de juros e impostos		$187.000
Juros pagos		14.000
Lucro tributável		$173.000
Impostos		60.550
Lucro líquido		$112.450
Dividendos	$33.735	
Acréscimo aos lucros retidos	78.715	

TURISMO DA ONÇA S/A — Balanço patrimonial de 31 de Dezembro de 2008

Ativo		Passivo e patrimônio líquido	
Ativo circulante		*Passivo circulante*	
Caixa	$ 25.300	Contas a pagar	$ 68.000
Contas a receber	40.700	Empréstimos a pagar	17.000
Estoque	86.900	Total	$ 85.000
Total	$152.900	Passivo não circulante	$158.000
Ativo imobilizado		*Patrimônio líquido*	
Instalações e equipamentos	413.000	Capital social e ágio	$140.000
		Lucros retidos	182.900
		Total	$322.900
Total do ativo	$565.900	Total do passivo e do patrimônio líquido	$565.900

27. **Cálculo da NAF** [OA2] No Problema 25, suponha que a empresa queira manter seu índice dívida/capital próprio constante. Qual é a NAF agora?

"A equação no texto": Ross §4.4, p. 109 (PDF 133), and Tabela 4.9, p. 111 (PDF 135):

$$\text{Taxa de crescimento interna} = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b} \qquad [4.2]$$

$$\text{Taxa de crescimento sustentável} = \frac{\text{ROE} \times b}{1 - \text{ROE} \times b} \qquad [4.3]$$

onde ROA = Lucro líquido / Total do ativo; ROE = Lucro líquido/Total do patrimônio líquido; b = Taxa de retenção = Acréscimo aos lucros retidos/Lucro líquido.

What is asked. (4.28) Redo 4.25 at 15% and 25% growth, plot NAF against g, find the g where NAF $= 0$, and explain why it differs from the internal growth formula. (4.29) The same for 4.27 at 20, 30 and 35%.

Approach. Write NAF as an explicit function of g using the model of Exercise 11: assets and payables scale with $1+g$; interest is fixed at 14,000; retained earnings $= (1-0.30)(1-0.35)\big[187{,}000(1+g) - 14{,}000\big] = 0.455\big[187{,}000(1+g) - 14{,}000\big]$. NAF is then linear in g, and its root is the model's own internal (4.28) or sustainable (4.29) rate.

Notation.

- $RE(g)$: addition to retained earnings at growth g; $D/E = 0.752555$ (total liabilities/equity).

Step 1 — (4.28) NAF(g) with no new debt or equity.

$$\begin{aligned}\text{NAF}(g) &= 565{,}900(1+g) - \left[68{,}000(1+g) + 17{,}000 + 158{,}000 + 322{,}900 + RE(g)\right] \\ &= (565{,}900 - 68{,}000 - 0.455 \cdot 187{,}000)(1+g) - (497{,}900 - 0.455 \cdot 14{,}000) \\ &= 412{,}815(1+g) - 491{,}530. \end{aligned} \tag{37}$$

At $g = 15\%$: $RE = 91{,}477.75$, NAF $= -16{,}792.75$; at 20%: $RE = 95{,}732$, NAF $= 3{,}848$; at 25%: $RE = 99{,}986.25$, NAF $= 24{,}488.75$. Root: $1 + g = 491{,}530/412{,}815$, so $g = 19.07\%$.

Step 2 — (4.28) The textbook formula. ROA $= 112{,}450/565{,}900 = 19.87\%$, ROA$\cdot b = 0.139097$, $g_{int} = 0.139097/(1-0.139097) = 16.16\%$. The formula equals the root of $565{,}900g - 78{,}715(1+g) = 0$, i.e. it assumes (i) no spontaneous liabilities and (ii) a constant net margin. Adding (i) back (payables grow with sales) moves the root to $78{,}715/(565{,}900 - 68{,}000 - 78{,}715) = 18.78\%$; adding (ii) (fixed interest makes the margin rise with sales) moves it to 19.07%.

Answer (4.28). NAF $= -16{,}793$; 3,848; 24,489. NAF $= 0$ at $g \approx 19.07\%$, above the formula's 16.16% because payables finance part of the growth and fixed interest raises the margin.

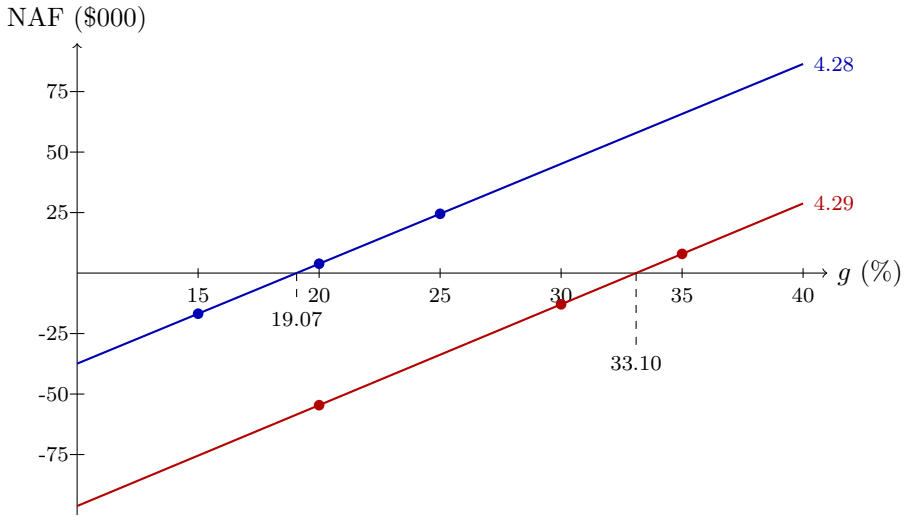
Step 3 — (4.29) NAF(g) with constant D/E. Sources are equity times $(1 + D/E)$:

$$\begin{aligned}\text{NAF}(g) &= 565{,}900(1+g) - 1.752555\left[322{,}900 + RE(g)\right] \\ &= (565{,}900 - 1.752555 \cdot 85{,}085)(1+g) - 1.752555(322{,}900 - 6{,}370) \\ &= 416{,}783.86(1+g) - 554{,}736.22. \end{aligned} \tag{38}$$

At 20%: NAF $= -54{,}595.59$; at 30% ($RE = 104{,}240.5$): $-12{,}917.21$; at 35% ($RE = 108{,}494.75$): 7,921.99. Root: $g = 554{,}736.22/416{,}783.86 - 1 = 33.10\%$.

Step 4 — (4.29) The textbook formula. ROE $= 112{,}450/322{,}900 = 34.83\%$, ROE $\cdot\, b = 0.243775$, $g_{sus} = 0.243775/(1 - 0.243775) = 32.24\%$, the root of $565{,}900g = 1.752555 \times 78{,}715(1+g) = 137{,}952.36(1+g)$, which assumes a constant margin. Payables do not matter here (they count as debt in D/E); the gap is the fixed interest only.

Answer (4.29). NAF $= -54{,}596$; $-12{,}917$; 7,922. NAF $= 0$ at $g \approx 33.10\%$, above the formula's 32.24% because interest does not grow with sales, so the margin rises with growth.



Intuition. NAF rises linearly with growth: each extra point of growth needs \$4,128 more financing in 4.28 (412,815/100) and \$4,168 in 4.29. The closed-form growth rates are exact only in the stylized world where every cost scales with sales; real pro formas have fixed costs (here interest), so always verify with the model itself.

Common mistakes.

- Treating the formula rates as the model's break-even rates without checking its assumptions.
- Plotting NAF against g with curvature: in this model it is a straight line.

Ref: Ross §4.4, "Necessidade de aporte financeiro e crescimento", pp. 106–111.

15. KMS Designs under 0.25% share growth — Berk 18.20–18.22 (18.12–18.15), pp. 560–561

Statement.

Source: Berk, Ch. 18, Exercises 18.20–18.22, p. 561 (PDF 592); Part 0 = Exercises 18.12–18.15, pp. 560–561 (PDF 591–592) · Answer in book: no

20. Forecast KMS's free cash flows (reproduce Table 18.13), assuming KMS's market share will increase by 0.25% per year; investment, financing, and depreciation will be adjusted accordingly; and working capital will be as you projected in Problem 15).
21. Calculate the continuation value of KMS using your reproduction of Table 18.8 from Problem 14, and assuming an EBITDA multiple of 8.5.
22. Assuming a cost of capital of 10%, compute the value of KMS under the 0.25% growth scenario.

Context needed to solve it **Part 0** — *Source: Berk, Ch. 18, Exercises 18.12–18.15, pp. 560–561 (PDF 591–592) · Answer in book: no.* (The book adds "For problems in this section, you should download the KMS spreadsheets available on the textbook's Web site"; the chapter tables copied below contain every number those spreadsheets hold.)

12. Assume that KMS's market share will increase by 0.25% per year rather than the 1% used in the chapter (see Table 18.5) and that its prices remain as in the chapter. What production capacity will KMS require each year? When will an expansion become necessary (that is, when will production volume exceed 1100)?
13. Under the assumption that KMS's market share will increase by 0.25% per year, you determine that the plant will require an expansion in 2012. The expansion will cost $20 million. Assuming that the financing of the expansion will be delayed accordingly, calculate the projected interest payments and the amount of the projected interest tax shields (assuming that KMS still uses a ten-year bond and interest rates remain the same as in the chapter) through 2015.
14. Under the assumption that KMS's market share will increase by 0.25% per year (and the investment and financing will be adjusted as described in Problem 13), you project the following depreciation:

Year	2010	2011	2012	2013	2014	2015
Depreciation	5,492	5,443	7,398	7,459	7,513	7,561

Using this information, project net income through 2015 (that is, reproduce Table 18.8 under the new assumptions).

15. Assuming that KMS's market share will increase by 0.25% per year (implying that the investment, financing, and depreciation will be adjusted as described in Problems 13 and 14), and that the working capital assumptions used in the chapter still hold, calculate KMS's working capital requirements through 2015 (that is, reproduce Table 18.9 under the new assumptions).

The chapter's KMS model, Berk §18.3–18.5, pp. 543–555 (PDF 574–586).

p. 543: KMS currently has the capacity to produce a maximum of 1.1 million units (i.e., 1100 thousand units). However, as detailed in Table 18.5, KMS expects both the total market size and its share of the market to grow to the point where the company will quickly exceed that capacity. Thus, KMS is considering an expansion that will increase its capacity to 2 million units—enough to handle its projected requirements through 2015.

Table 18.5, "KMS's Forecasted Production Capacity Requirements", p. 543:

Year	2010	2011	2012	2013	2014	2015
Market Size (000s units)	10,000	10,500	11,025	11,576	12,155	12,763
Market Share	10.0%	11.0%	12.0%	13.0%	14.0%	15.0%
Production Volume (Line 3 × Line 4)	1,000	1,155	1,323	1,505	1,702	1,914
Average Sales Price	$ 74.89	$ 76.51	$ 78.04	$ 79.60	$ 81.19	$ 82.82

pp. 543–544: The new equipment to increase KMS's capacity will cost $20 million and will need to be purchased in 2011 to meet the company's production needs. [. . .] KMS has ongoing capital investment requirements to cover the replacement of existing equipment—these were expected to be $5 million per year without the new equipment. The additional $20 million is reflected in 2011, bringing the total to $25 million for 2011 and increasing expected recurring investment to $8 million per year in years 2012–2015.

Table 18.6, "KMS's Forecasted Capital Expenditures" ($000s), p. 543:

Year	2010	2011	2012	2013	2014	2015
Opening Book Value	49,919	49,427	66,984	67,486	67,937	68,344
Capital Investment	5,000	25,000	8,000	8,000	8,000	8,000
Depreciation	−5,492	−7,443	−7,498	−7,549	−7,594	−7,634
Closing Book Value	49,427	66,984	67,486	67,937	68,344	68,709

p. 544: KMS plans to finance the new equipment by issuing ten-year coupon bonds with a coupon rate of 6.8%. Thus, KMS will pay only interest on the bonds until the repayment of principal in ten years. The principal on its outstanding debt of $4500 is also not due before 2015.

Table 18.7, "KMS's Planned Debt and Interest Payments" ($000s), p. 544 (rate 6.80%):

Year	2010	2011	2012	2013	2014	2015
Outstanding Debt	4,500	24,500	24,500	24,500	24,500	24,500
Net New Borrowing	—	20,000	—	—	—	—
Interest on Debt	306	306	1,666	1,666	1,666	1,666

$$\text{Interest in Year } t = \text{Interest Rate} \times \text{Ending Balance in Year } (t-1) \qquad (18.1)$$

$$\text{Sales} = \text{Market Size} \times \text{Market Share} \times \text{Average Sales Price} \qquad (18.2)$$

p. 544: We will assume that costs except depreciation will continue to be 78% of sales [. . .]. *p. 545:* KMS pays a 35% tax rate, and the income tax is computed as: Income Tax = Pretax Income × Tax Rate (18.3).

Table 18.8, "Pro Forma Income Statement for KMS Expansion" ($000s), p. 545:

Year	2010	2011	2012	2013	2014	2015
Sales	74,889	88,369	103,247	119,793	138,167	158,546
Costs Except Depreciation	−58,413	−68,928	−80,533	−93,438	−107,770	−123,666
EBITDA	16,476	19,441	22,714	26,354	30,397	34,880
Depreciation	−5,492	−7,443	−7,498	−7,549	−7,594	−7,634
EBIT	10,984	11,998	15,216	18,806	22,803	27,246
Interest Expense (net)	−306	−306	−1,666	−1,666	−1,666	−1,666
Pretax Income	10,678	11,692	13,550	17,140	21,137	25,580
Income Tax	−3,737	−4,092	−4,742	−5,999	−7,398	−8,953
Net Income	6,941	7,600	8,807	11,141	13,739	16,627

p. 545: We have forecast that the minimum required cash will be 16% of sales, accounts receivable will be 19% of sales, inventory will be 20% of sales, and accounts payable will be 16% of sales, all as they were in 2010.

Table 18.9, "KMS Projected Working Capital Needs" ($000s), p. 545:

Year	2010	2011	2012	2013	2014	2015
Cash	11,982	14,139	16,520	19,167	22,107	25,367
Accounts Receivable	14,229	16,790	19,617	22,761	26,252	30,124
Inventory	14,978	17,674	20,649	23,959	27,633	31,709
Total Current Assets	41,189	48,603	56,786	65,886	75,992	87,201
Accounts Payable	11,982	14,139	16,520	19,167	22,107	25,367
Total Current Liabilities	11,982	14,139	16,520	19,167	22,107	25,367
Net Working Capital (7 − 10)	29,207	34,464	40,266	46,719	53,885	61,833
Increase in Net Working Capital		5,257	5,802	6,453	7,166	7,948

pp. 551–552 (PDF 582–583): To compute KMS's free cash flows, which exclude cash flows associated with leverage, we first adjust net income by adding back the after-tax interest payments associated with the net debt in its capital structure. [...] After-Tax Interest Expense = (1 − Tax Rate) × (Interest on Debt) − (1 − Tax Rate) × (Interest on Excess Cash) (18.6). [...] To compute KMS's free cash flows from its unlevered net income, we add back depreciation (which is not a cash expense), and deduct KMS's increases in net working capital and capital expenditures.

Table 18.13, "KMS Forecasted Free Cash Flows" ($000s), p. 551:

Year	2011	2012	2013	2014	2015
Net Income	7,600	8,807	11,141	13,739	16,627
Plus: After-Tax Interest Expense	199	1,083	1,083	1,083	1,083
Unlevered Net Income	7,799	9,890	12,224	14,822	17,710
Plus: Depreciation	7,443	7,498	7,549	7,594	7,634
Less: Increases in NWC	(5,257)	(5,802)	(6,453)	(7,166)	(7,948)
Less: Capital Expenditures	(25,000)	(8,000)	(8,000)	(8,000)	(8,000)
Free Cash Flow of Firm	(15,015)	3,586	5,320	7,250	9,396

pp. 553–555 (PDF 584–586): As we learned in Chapter 16, absent distress costs, the value of a firm with debt is equal to the value of the firm without debt plus the present value of its interest tax shields. [...] we compute the present value of the unlevered free cash flows of KMS and add to it the present value of the tax shields created by our planned interest payments. [...] Continuation Enterprise Value at Forecast Horizon = EBITDA at Horizon × EBITDA Multiple at Horizon (18.7). [...] Assume that KMS's financial managers have estimated KMS's unlevered cost of capital to be 10% (specifically, 10% is their pretax WACC [...]). [...] we discount KMS's free cash flows at the firm's pretax WACC of 10%. [...] Interest Tax Shield = Net Interest Expense × Tax Rate (18.10). [...] we calculate the present value of the interest tax shield using the interest rate on debt as the discount rate, not using the WACC. Recall that the reason for doing so is that the tax shield is only as risky as the debt that creates it, so that the proper discount rate is the debt's interest rate, here at 6.8% [...]. Footnote 14: We have not ignored the rest of the interest tax shields from the new debt. The value of those shields is subsumed in the continuation value of the firm.

Table 18.14, "Calculation of KMS Firm Value with the Expansion" ($000s), p. 554 (PDF 585):

Year	2010	2011	2012	2013	2014	2015
Free Cash Flow of Firm		(15,015)	3,586	5,320	7,250	9,396
PV Free Cash Flow (at 10%)	4,096					
Continuation Value						313,920
PV Continuation Value (at 10%)	194,920					
Net Interest Expense		306	1,666	1,666	1,666	1,666
Interest Tax Shield		107	583	583	583	583
PV Interest Tax Shield (at 6.8%)	1,958					
Firm Value (3 + 5 + 8)	200,974					

(In the chapter, 2015 EBITDA of $34.880 million × a multiple of 9 = $313.920 million.)

What is asked. (18.12) Required production each year and when it first exceeds the 1,100 thousand-unit capacity. (18.13) Interest and interest tax shields through 2015 when the $20 million expansion

is delayed to 2012. (18.14) Net income through 2015 with the given depreciation. (18.15) Working capital needs. (18.20) Free cash flows. (18.21) Continuation value at 8.5 times 2015 EBITDA. (18.22) Firm value at a 10% cost of capital.

Approach. Rebuild the KMS model (Tables 18.5–18.9, 18.13–18.14) with market share $10\% + 0.25\% \times t$. Value with APV, as in the chapter: unlevered free cash flows and the continuation value at the 10% pretax WACC, plus interest tax shields at the 6.8% debt rate (later tax shields are inside the multiple). Assumption, stated because the statement says "adjusted accordingly": capex is \$5 million in 2011 (replacement only), \$25 million in 2012 (expansion year) and \$8 million from 2013, mirroring the chapter's schedule one year later.

Notation.

- $Q_t = M_t \times s_t$: production (000 units); $S_t = Q_t \times p_t$: sales (\$000); costs 78% of sales; $\tau = 35\%$.
- NWC_t: cash 16% + receivables 19% + inventory 20% − payables 16% of sales = 39% of sales (2010 actual: 29,207).
- $FCF_t = NI_t + (1-\tau)Int_t + Dep_t - \Delta NWC_t - CapEx_t$; $CV_{2015} = 8.5 \times EBITDA_{2015}$.

Step 1 — (18.12) Capacity. Volumes $Q_t = M_t s_t$ (e.g. 2012: $11{,}025 \times 10.5\% = 1{,}157.625$):

	2010	2011	2012	2013	2014	2015
Market share (%)	10.00	10.25	10.50	10.75	11.00	11.25
Production (000 units)	1,000.00	1,076.25	1,157.63	1,244.42	1,337.05	1,435.84

Answer (18.12). Requirements as in the table; production first exceeds 1,100 in 2012, so the expansion is needed in 2012.

Step 2 — (18.13) Interest and tax shields. By Eq. (18.1), interest in year t is 6.8% of debt at the end of $t-1$. Debt is 4,500 through 2011 and 24,500 from the end of 2012: $0.068(4{,}500) = 306$ in 2011 and 2012; $0.068(24{,}500) = 1{,}666$ in 2013–2015. Tax shields = 0.35× interest: 107.1, 107.1, 583.1, 583.1, 583.1.

Answer (18.13). Interest 306, 306, 1,666, 1,666, 1,666; tax shields 107.1, 107.1, 583.1, 583.1, 583.1 (\$000).

Step 3 — (18.14) Pro forma income statement. Sales = $Q_t p_t$ (2011: $1{,}076.25 \times 76.51 = 82{,}343.9$); then costs, EBITDA, depreciation (given), interest (Step 2), tax 35%:

\$000	2010	2011	2012	2013	2014	2015
Sales	74,889.0	82,343.9	90,341.1	99,055.8	108,555.1	118,916.1
Costs except dep. (78%)	−58,413.0	−64,228.2	−70,466.0	−77,263.5	−84,673.0	−92,754.5
EBITDA	16,476.0	18,115.7	19,875.0	21,792.3	23,882.1	26,161.5
Depreciation	−5,492.0	−5,443.0	−7,398.0	−7,459.0	−7,513.0	−7,561.0
EBIT	10,984.0	12,672.7	12,477.0	14,333.3	16,369.1	18,600.5
Interest	−306.0	−306.0	−306.0	−1,666.0	−1,666.0	−1,666.0
Pretax income	10,678.0	12,366.7	12,171.0	12,667.3	14,703.1	16,934.5
Income tax (35%)	−3,737.3	−4,328.3	−4,259.9	−4,433.5	−5,146.1	−5,927.1
Net income	6,940.7	8,038.3	7,911.2	8,233.7	9,557.0	11,007.4

Answer (18.14). Net income 8,038.3; 7,911.2; 8,233.7; 9,557.0; 11,007.4 for 2011–2015 (\$000).

Step 4 — (18.15) Working capital. Each item is its percent of sales (2010 actual):

\$000	2010	2011	2012	2013	2014	2015
Cash (16%)	11,982.0	13,175.0	14,454.6	15,848.9	17,368.8	19,026.6
Accounts receivable (19%)	14,229.0	15,645.3	17,164.8	18,820.6	20,625.5	22,594.1
Inventory (20%)	14,978.0	16,468.8	18,068.2	19,811.2	21,711.0	23,783.2
Total current assets	41,189.0	45,289.1	49,687.6	54,480.7	59,705.3	65,403.8
Accounts payable (16%)	11,982.0	13,175.0	14,454.6	15,848.9	17,368.8	19,026.6
Net working capital	29,207.0	32,114.1	35,233.0	38,631.8	42,336.5	46,377.3
Increase in NWC		2,907.1	3,118.9	3,398.8	3,704.7	4,040.8

Answer (18.15). NWC as in the table; increases 2,907.1; 3,118.9; 3,398.8; 3,704.7; 4,040.8 ($000).

Step 5 — (18.20) Free cash flows. Unlevered net income $= NI + (1 - 0.35)Int$ $(= 0.65\,EBIT$; after-tax interest is 198.9 while debt is 4,500 and 1,082.9 after). E.g. 2012: $8{,}110.1 + 7{,}398 - 3{,}118.9 - 25{,}000 = -12{,}610.8$.

$000	2011	2012	2013	2014	2015
Net income	8,038.3	7,911.2	8,233.7	9,557.0	11,007.4
Plus: after-tax interest	198.9	198.9	1,082.9	1,082.9	1,082.9
Unlevered net income	8,237.2	8,110.1	9,316.6	10,639.9	12,090.3
Plus: depreciation	5,443.0	7,398.0	7,459.0	7,513.0	7,561.0
Less: increase in NWC	−2,907.1	−3,118.9	−3,398.8	−3,704.7	−4,040.8
Less: capital expenditures	−5,000.0	−25,000.0	−8,000.0	−8,000.0	−8,000.0
Free cash flow of firm	5,773.1	−12,610.8	5,376.9	6,448.2	7,610.6

Answer (18.20). FCF = 5,773.1; −12,610.8; 5,376.9; 6,448.2; 7,610.6 ($000).

Step 6 — (18.21) Continuation value. By Eq. (18.7),

$$CV_{2015} = 8.5 \times EBITDA_{2015} = 8.5(26{,}161.5) = 222{,}373.0. \tag{39}$$

Answer (18.21). About $222.4 million at the end of 2015.

Step 7 — (18.22) APV. Discount FCF and CV at 10% ($1.1^5 = 1.61051$) and tax shields at 6.8%:

$$\begin{aligned}
PV(FCF) &= \frac{5{,}773.1}{1.1} + \frac{-12{,}610.8}{1.1^2} + \frac{5{,}376.9}{1.1^3} + \frac{6{,}448.2}{1.1^4} + \frac{7{,}610.6}{1.1^5} \\
&= 5{,}248.3 - 10{,}422.2 + 4{,}039.7 + 4{,}404.2 + 4{,}725.6 = 7{,}995.6, \\
PV(CV) &= 222{,}373.0/1.61051 = 138{,}076.2, \\
PV(ITS) &= \frac{107.1}{1.068} + \frac{107.1}{1.068^2} + \frac{583.1}{1.068^3} + \frac{583.1}{1.068^4} + \frac{583.1}{1.068^5} \\
&= 100.3 + 93.9 + 478.7 + 448.2 + 419.6 = 1{,}540.7, \\
V_{2010} &= 7{,}995.6 + 138{,}076.2 + 1{,}540.7 = 147{,}612.5.
\end{aligned} \tag{40}$$

Answer (18.22). KMS is worth about $147.6 million under the 0.25% scenario, versus $201.0 million in the chapter's 1% scenario (Table 18.14) — $53.4 million less (still $42.5 million less if the chapter's value is redone at 8.5×: $190.1 million).

Intuition. The value lives in the continuation value (93.5% of the total), and the continuation value lives in 2015 EBITDA. Slower share growth cuts 2015 EBITDA from 34,880 to 26,162, and with it most of the value; the five-year cash flows barely matter. That is the chapter's lesson: growth is valuable only through the cash flows it adds, and the forecast's horizon multiple must be stress-tested.

Common mistakes.

- Subtracting the level of NWC instead of its *increase* (Berk's "Common Mistake", p. 553).
- Discounting the tax shields at 10%, or adding them twice (net income already contains them; start from *unlevered* income).
- Charging the new bond's interest in 2012: under Eq. (18.1) interest starts the year after the borrowing.

Ref: Berk §18.3–18.5, pp. 542–556; Tables 18.5–18.9, 18.13–18.14.